Lieb-Robinson bounds from one theorem

A Lieb-Robinson bound says that in a lattice of quantum systems with bounded local interactions, information travels no faster than a finite velocity: the commutator of a time-evolved observable with a distant one is exponentially small outside a light cone. These notes are the mathematical account of the Lean library LocalityLean. They prove one theorem, the path-sum bound: under three hypotheses on the generators of the dynamics, it bounds such commutators by a sum over paths of overlapping interaction terms. We then read off the published bounds for qudit lattices and open systems as its instances, and obtain the bound for harmonic oscillators from the combinatorics of its closed forms; the chapters on fermions and infinite lattices are to come.

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About these notes

We assume familiarity with C*-algebras, bounded operators on a Banach space and their exponential series, and, for Chapter 7, the Weyl form of the canonical commutation relations. After the notation come a reading guide, which translates the generators and paths used here into the interactions and graphs of the literature, and a roadmap of the chapters; an index of terms precedes the references.

A statement marked Lean ✓ has a checked proof in LocalityLean or in the libraries LinearAQFT, LocalAlgebrasLean and BosonicLean on which it builds, whose correspondence with the statement here has been independently reviewed; the declaration index at the end names the checked declarations. Every Example, and the one Proposition without a Lean proof (on the rescaling of units for harmonic bosons), carries exactly one status: Lean ✓ as above, script ✓ when a check script verifies it exhaustively, or paper when it is checked on paper only; the scripts behind the numbers are named in the Checks paragraph that closes each chapter.

Notation

A letter that serves more than one purpose is listed with each use.

Symbol	Meaning
Λ	the set of sites: any set in Chapters 1–3 (finite where the matrix form needs it); a finite set in Chapters 4–5
$X, Y \subseteq \Lambda$	regions; X carries the observable a, Y the probe b
$\mathcal{A}$	the ambient C*-algebra; $\mathcal{A}(X)$ the local algebra of X (a net, monotone in X)
$[a, b]$	the commutator $ab - ba$; $[a, b]_\pm$ the graded bracket (Chapters 6–8, to come)
I, $i \in I$	the finite index set of the generator family
$S_i \subseteq \Lambda$	the support of the i-th generator, a finite set
L_i, $L = \sum_i L_i$	the local generators and their sum, bounded linear maps of $\mathcal{A}$
$I^{\partial Y}$	the terms whose support crosses Y (meets Y and its complement)
$L^{\partial Y}$, L'	the generator of the crossing terms, and that of the rest, the terms disjoint from Y or inside Y (L' depends on Y)
$\kappa_i \geq 0$, $K = \sum_i \kappa_i$	the strength of the i-th generator (hypothesis G1) and the total strength

Symbol	Meaning
$J \subseteq I$, L_J	a sub-family of indices and its generator $\sum_{i\in J} L_i$
$N_S(c)$	the commutator seminorm $\sup\{\|[c,b]\| : b \in \mathcal{A}(S), \|b\| \le 1\}$ of $c \in \mathcal{A}$ relative to the region S
$W, Z \subseteq \Lambda$	further regions; Z in definitions, W in the support lemma
a, b, c	elements of $\mathcal{A}$: the observable, the probe, and a general element
s	an intermediate time, $0 \le s \le t$
$A(t) = e^{tL}a$, $B(t) = e^{tL}b$	the evolved observable and probe, $t \ge 0$ in Chapters 2–3 and all real t for a contractive negation; $e^{\mathrm{i}tH}ae^{-\mathrm{i}tH}$ in the Hamiltonian case
f; f_0, f_1, f_2	an auxiliary function inside a proof; in the sources' route, $f(t) = [\tau_t(A), B]$ with the sources' capital letters; the coefficients of the evolved observable in Example 2.11
$\sigma^x, \sigma^y, \sigma^z$; σ^x_u	the Pauli matrices (written out in Chapter 4); σ^x at the site u
$C_{X,Y}(t)$	$\sup\{\|[e^{tL}a, b]\| : a \in \mathcal{A}(X), b \in \mathcal{A}(Y), \|a\|, \|b\| \le 1\}$
$\mathbf{1}_{\{P\}}$	the indicator of a condition P: 1 if it holds and 0 otherwise (Chapter 2)
β_i	a continuous majorant of $N_{S_i}(A(s))$ in Theorem 2.7
$\pi = (i_1, \dots, i_k)$	a path of length k from Y to X; $w(\pi) = \prod_j \kappa_{i_j}$ its weight
$P_k(Y, X)$	$\sum_\pi w(\pi)$ over the paths of length k
$\gamma_k(t)$, $B_n(Y, X, t)$	the coefficient $(2t)^{k+1}/(k+1)!$ and the truncated bound of Theorem 2.8
$\mathbb{B}$; R, Q, G	the Banach algebra of Duhamel's identity (a complete normed unital real algebra); elements of it in the identity and its proof
E, M	a Banach space and a closed subspace in the invariance lemma
H, Φ, τ_t	a Hamiltonian, the sources' interaction, and their Heisenberg dynamics; in Chapter 4, $H = \sum_i \Phi_i$
$h = (h_{uu'})$	the site matrix, $h_{uu'} = \sum_{i:\, u,u' \in S_i} \kappa_i$ (Chapter 3)
T; ∂Y	the touched sites $\bigcup_i S_i$; the boundary of Y, its sites touched by a term crossing Y
$\mathrm{d}(\partial Y, X \cap T)$	the smallest distance $d(y, x)$ from a site y of the boundary ∂Y to a site x of $X \cap T$ (Chapter 3)
$w_k(y, x)$, $W_k(Y, X)$; $w_k(z, X)$	the walk weights and their sum over $\partial Y \times (X \cap T)$; the sum of $w_k(z, x)$ over $x \in X \cap T$ (Chapter 3)
Q; r	in Chapter 3: the terms crossing Y and missing X; the number of steps a walk needs to reach X
F_0, T'	a decay function of the distance, and a finite set of sites in the reproducing property
K_1	$\sum_i \kappa_i \lvert S_i \rvert$, a bound on the row sums of h
N_F, $\Sigma_F(Y, X)$	an F-bound of the family ($h \le N_F F$) and the sum of F over $\partial Y \times (X \cap T)$
g, ℓ, p, ρ	bounded geometry: terms per site, strength, support size, support diameter
$\chi_i(u)$	the indicator $\mathbf{1}_{\{u \in S_i\}}$ of a support
$d(u, u')$; q	the distance of a bounded geometry ($d(u,u) = 0$ and the triangle inequality); a bound on the row sums of h
λ	the decay rate of the decay function in the chain example (Chapter 3)
F, C_F, $\|\Phi\|_F$	a decay function, its reproducing constant, the F-norm of an interaction (Chapter 3)
v, μ	the velocity $2g\ell p$ of the bounded-geometry bound and the decay rate of the light cone (Chapter 3)

Symbol	Meaning
x, y	sites (Chapters 3–5); x also a nonnegative real in Lemma 3.12 and the proof of Theorem 3.14; in the first half of Chapter 4 also elements of $\mathcal{A}$, in L_x, R_y, ad_x
L_x, R_x	the left and right multiplications by x (Chapter 4)
ad_x	the derivation $a \mapsto \mathrm{i}[x, a]$
Φ_i, H_J, U_t	the terms of a Hamiltonian family, the Hamiltonian of a sub-family, and the unitary $e^{\mathrm{i}tH}$
Ω_u, Ω; n	the alphabet at the site u, a uniform alphabet; the dimension of a qudit, $\Omega_u = \{0, \ldots, n-1\}$ (n is also the length of a chain in the examples)
s, $s \sim_X s'$, $s\vert_X$	a configuration; agreement off X; the restriction to X
r, r'	configurations, in Chapter 4
M, N	matrices of the qudit algebra (Chapters 4–5)
m; m_i	a matrix on the letters of a region; the matrices of a qudit interaction (Chapter 4)
ξ, η; $(\xi, s\vert_{X^c})$	choices of letters on a region (X in Chapter 4, the complement $Y = X^c$ in Chapter 5); the configuration with the letters ξ on X and those of s off X
$(s\vert_X, \eta)$	the configuration with the letters of s on X and η off X (Chapter 5)
l, l'	letters of one cyclic alphabet $\mathbb{Z}_n$, indexing the clock and shift matrices and the character sum (Chapter 5)
q_u	the order of the cyclic alphabet $\Omega_u = \mathbb{Z}_{q_u}$ at the site u (Chapter 5)
$\omega_k(t)$	the walk term $(2t)^{k+1} W_{k+1}(Y, X)/(k+1)!$ of the pair (Y, X), $k \geq 0$ (Chapter 3, used in Chapter 5)
ζ; Θ	re-indexed letters in the twirl proof; the conjugation $a \mapsto \Gamma a \Gamma$ of the dephasing proof (Chapter 5)
α; $D(X, Y)$	the exponent of the polynomial decay function $F_0(r) = (1+r)^{-\alpha}$; the smaller of the two boundary sums in the sources' bound (Chapter 4)
M_Λ	the qudit algebra, the matrices indexed by configurations
$\iota_X m$	the embedding of a matrix on the letters of X (the sources' $m \otimes 1$)
m_u	the single-site operator, m at the site u and the identity elsewhere
M^φ, F_μ	the relabelling of a matrix by a bijection φ of the sites; the weighted decay function $e^{-\mu d} F$
$\Vert F \Vert$	$\sup_x \sum_y F(x, y)$, the uniform integrability of a decay function (Chapter 4)
ν	the dimension of the lattice $\mathbb{Z}^\nu$ (Chapters 3–4)
θ	the parity automorphism of a graded net (Chapters 6–8, to come)
P_X	the conditional expectation onto $\mathcal{A}(X)$, the normalized partial trace over X^c re-embedded, equal to the finite twirl (Chapter 5)
Ω_Y, $\vert\Omega_Y\vert$	the letters on a region Y and their number (Chapter 5)
$e_n(k)$	the character $e^{2\pi \mathrm{i} k/n}$ of $\mathbb{Z}_n$ (Chapter 5)
V, Λ', $\bar{a}$	the phase space $\Lambda \to \mathbb{R}^2$ of harmonic bosons, its coordinates $\Lambda \times \{q, p\}$, the other coordinate at the site of a (Chapter 7)
σ, Ω	the symplectic form $\sum_x (f_x^q g_x^p - f_x^p g_x^q) = f \cdot \Omega g$ and its matrix (Chapter 7); Ω_u is the alphabet of a site in Chapter 4
$W(f)$	the Weyl operator of $f \in V$, with $W(f)W(g) = e^{-\mathrm{i}\sigma(f,g)/2} W(f+g)$ (Chapter 7)

Symbol	Meaning
K, T_t, α_t	a quadratic generator ($K^{\mathsf{T}}\Omega + \Omega K = 0$), its flow e^{tK} and the dynamics $W(f) \mapsto W(T_t f)$ (Chapter 7); K is the total strength in Chapter 2
$\lvert K\rvert$, R, v	the entrywise absolute value of a matrix, the range and the row sum of a generator (Chapter 7)
Z, X	the clock and shift matrices of a cyclic alphabet (Chapter 5)
ψ_η, $V_{\xi,\eta}$	the clock phase of the letters η and the Weyl unitary of the letters ξ, η on a region (Chapter 5)
$\mathcal{T}$, $\mathcal{N}$, ε	a map of $\mathcal{A}$, a neighbourhood map of regions, and the error of ε-locality (Chapter 5)
$\varepsilon(t)$, $\varepsilon_r(t)$	the errors of ε-locality of e^{tL}, for a neighbourhood map $\mathcal{N}$ and for the r-neighbourhood (Chapter 5)
X'	the larger region $X' \supseteq X$ of the truncated dynamics, with $Y = X'^c$ (Chapter 5)
O, O^{+r}	a region carrying an observable in the locality condition, and its r-neighbourhood (Chapter 5)
Γ_j, Γ_{ij}, Δ_Γ, L_S	the jump operators of a Lindblad term and those of the i-th term of a Lindblad family, the dissipator of one jump, and the generator of the term (Chapter 5)

Three languages for one bound: a reading guide

Lieb-Robinson bounds are written in two dialects. The review [1] speaks of a graph whose edges carry Hamiltonian terms H_e, of a *site matrix* h_{uv} adding up the strengths of the terms containing two sites, and of walks on the graph; the mathematical-physics papers [4, 6] speak of an interaction Φ assigning a self-adjoint term $\Phi(Z)$ to each finite set Z of sites, of a *decay function* F with a reproducing property, and of the F-norm $\|\Phi\|_F$ (Chapter 3 defines the site matrix and the decay functions).

These notes use a third, slightly more general language, chosen so that one theorem covers Hamiltonian and open dynamics alike: a finite family of *local generators* L_i with supports S_i and strengths κ_i, and *paths* of supports from the probe's region to the observable's. The table says which object plays which role, and where each conversion is proved.

Review (graph language)	Mathematical physics (interactions)	These notes (generators)
edge Hamiltonian H_e, $e = \{u, v\}$	interaction term $\Phi(Z)$, Z a finite set	generator L_i with support S_i; Hamiltonian case $L_i = \mathrm{i}[\Phi(S_i), \cdot]$ (Chapter 4)
strength $\|H_e\|$	$\|\Phi(Z)\|$	constant κ_i; equals $\|\Phi(S_i)\|$ in the Hamiltonian case; for a Lindblad term the direct strength $\|H\| + \sum_j \|\Gamma_j\|^2$, by a commutator identity, or $\|L_S\|$ through the conditional expectation (Chapters 4, 5)
Heisenberg evolution $e^{\mathrm{i}tH} A e^{-\mathrm{i}tH}$	$\tau_t(A)$, the automorphism group	$e^{tL}A$ with $L = \sum_i L_i$; hypotheses G1–G3 (Definition 1.4); equal to the conjugation in the Hamiltonian case (Chapter 4)

Review	Mathematical physics	These notes
operators on the qubits of a region	$\mathcal{A}_X = \bigotimes_{x\in X} \mathrm{M}_n \otimes 1$, the tensor factor	the matrices local to X: zero between configurations differing off X, entries depending on the letters inside X (Chapter 4)
bosons absent	Weyl operators $W(f)$ with $\tau_t(W(f)) = W(f_t)$, the kernel f_t by normal modes	a quadratic generator K on the Weyl labels, the flow $T_t = e^{tK}$ and $\alpha_t(W(f)) = W(T_t f)$; the site matrix is $\lvert K\rvert$ (Chapter 7)
$D(X,Y)$ absent	$D(X,Y) = \min\{\sum_{X\times\partial Y}, \sum_{\partial X\times Y}\}$, eq. (3.24) of [6]	the smaller of the two walk sums or sums of the decay function, by the role exchange of automorphisms (Chapter 4)
site matrix $h_{uv} = \sum_{e\ni u,v}\lVert H_e\rVert$, eq. (3.35)	$\sum_{Z\ni u,v}\lVert\Phi(Z)\rVert \le \lVert\Phi\rVert_F\, F(\mathrm{d}(u,v))$, eq. (3.14)	site matrix $h_{uv} = \sum_{i:\, u,v\in S_i} \kappa_i$, with diagonal $h_{uu} = \sum_{i:\, u\in S_i} \kappa_i$ (Chapter 3)
walk on the graph; irreducible path (Chen–Lucas)	chain of interaction supports with boundary-crossing sets, eqs. (3.41)–(3.42) of [6]	path: indices $i_1, \dots, i_k$ with S_{i_1} crossing Y and each later support crossing the previous one (no term follows itself, none is nested in its predecessor), with first contact with X at step k (Definition 1.7)
$\mathcal{C}_{X,Y}(t) \le \sum_{x\in X, y\in Y}(e^{2\lvert t\rvert h})_{xy}$, eq. (3.36), with $\mathcal{C} = \frac{1}{2}\sup\lVert[A(t),B]\rVert$ by its eq. (3.24) (its (3.22) omits the $\frac{1}{2}$, a misprint)	$\lVert[\tau_t(A),B]\rVert \le \frac{2\lVert A\rVert\lVert B\rVert}{C_F}(e^{2\lVert\Phi\rVert_F C_F t} - 1)D(X,Y)$, Thm. 3.1 of [6]	path series (Theorem 2.9), then the two closed forms as corollaries (Chapter 3); the matrix form is the review's (3.36) with its constant doubled and the -1 that the series proof gives
Haar average over unitaries on X^c, Prop. 4.1	conditional expectation $\mathbb{E}_X$ as a finite group average, eq. (4.6) of [5]	conditional expectation P_X, the normalized partial trace, computed as a finite twirl over the Weyl unitaries of X^c (Chapter 5); also a second route to G1 for Lindblad terms, with the strength $\lVert L_S\rVert$
$\lVert\mathbb{E}_S A - A\rVert \le \sup\lVert[A,B]\rVert \le 2\lVert\mathbb{E}_S A - A\rVert$, Prop. 4.1	Lemma 4.1 of [6]	local approximation, the same constants (Theorem 5.10); on any causal net the converse holds for every element of $\mathcal{A}(X)$
Truncated dynamics $\lVert\tau_t(A) - \tau_t^B(A)\rVert$ under the Hamiltonian restricted to B, Prop. 4.3	—	$\lVert A(t) - P_{X'}A(t)\rVert$ with the conditional expectation as the local approximant, a corollary of the walk bound with our constants (Theorem 5.11); stated as ε-locality of e^{tL}, the approximate form of a cellular automaton's locality (Corollary 5.12)

Review	Mathematical physics	These notes
Lindblad generator, Thm. 3.14, with the norms of the terms as strengths	—	the same theorem: a Lindblad family is a generator family satisfying G1–G3, G3 as the contraction hypothesis implied by the review's complete positivity (Chapter 5)
—	CAR algebra, parity automorphism, $[A, B] = 0$ or $\{A, B\} = 0$ for disjoint supports, Prop. 2.1 of [5]	graded net; graded bracket $[A, B]_{\pm}$ (Chapter 6)

Reconnections. Each chapter from Chapter 2 on ends with the sources' statements recovered in the sources' notation: the review's Theorem 3.4 in Chapter 2; Theorem 3.1 of [6] in Chapters 3 and 4; the review's bound for spin systems, its eq. (3.36), in Chapter 4; the review's Theorem 3.14 and Proposition 4.1 in Chapter 5; Theorem 3.1 of [7] in Chapter 7; Theorem 3.1 of [5] in the chapter to come. Chapter 2 sketches the route of the sources through the Jacobi identity.

Roadmap

The notes follow the logical build of the theory rather than the order of the literature: the objects (nets of local algebras, local generators, paths), the central theorem (the path-sum bound), its closed forms (site matrix, decay functions, light cones), then one chapter per instance, each showing the same theorem specialized, and finally the thermodynamic limit. Figure 1 shows which chapters use which. Chapter 7 is the exception to the pattern: the generator of harmonic bosons is unbounded, so the path-sum bound of Chapter 2 does not apply, and the chapter uses only the combinatorics of Chapter 3.

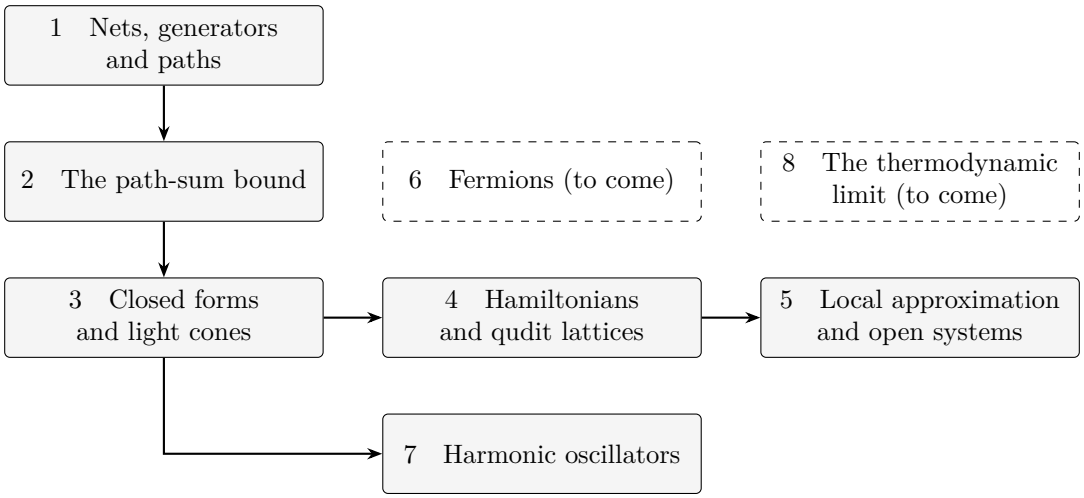


Figure 1: The chapters and their dependence: an arrow points from a chapter to one that uses it. Dashed: chapters to come.

1 Nets, generators and paths

A Lieb-Robinson bound is a statement about three things: where observables live, how the dynamics is assembled from local pieces, and which chains of pieces can carry a signal from one region to another. In this chapter we fix the three objects. The net of local algebras records where observables live (Definition 1.1); the local generator family records the dynamics as a finite sum of bounded maps, each tied to a patch of sites and each obeying three hypotheses (Definition 1.4); the paths record the chains (Definition 1.7).

The path-sum bound of Chapter 2 bounds the commutator of an observable a on a region X, evolved in time, with a second observable b, the *probe*, on a region Y, by a weighted count of paths from Y to X. A reader who knows the literature's interactions $\Phi(Z)$ will recognise the Hamiltonian case in Example 1.6.

1.1 The net of local algebras

Throughout, Λ is a set, the *sites*, and $\mathcal{A}$ a unital C*-algebra, the *ambient algebra*. The regions are all subsets of Λ.

Definition 1.1 (net of local algebras)**.** A *net* assigns to every region $Z \subseteq \Lambda$ a unital C*-subalgebra $\mathcal{A}(Z) \subseteq \mathcal{A}$ (a norm-closed $*$-subalgebra containing 1), the *local algebra* of Z, monotonically: $\mathcal{A}(Z) \subseteq \mathcal{A}(Z')$ whenever $Z \subseteq Z'$. The net is *causal* if observables of disjoint regions commute: $ab = ba$ for $a \in \mathcal{A}(Z)$, $b \in \mathcal{A}(Z')$ and $Z \cap Z' = \emptyset$.

The net is the one of algebraic quantum field theory, an assignment of local algebras to regions over an abstract causal structure that says which regions are spacelike separated and which depend on which. On a lattice the structure is the simplest one: two regions are spacelike separated exactly when they are disjoint, and a region depends on another exactly when it is contained in it. Causality of the net for this structure is the commutation above, and it is the only property of the structure used in these notes. We require the assignment on all subsets, including infinite ones, because the proof of the path-sum bound uses the local algebra of the complement Y^c of the probe's region, which is infinite when Λ is infinite and Y finite. On a finite lattice this is no restriction; on the infinite lattice of Chapter 8 the quasi-local algebra supplies $\mathcal{A}(Z)$ for every Z.

1.2 The commutator seminorm

A bounded map of $\mathcal{A}$ can only be called local if there is a way to say how much an observable *sees* a region. The right quantity is not the norm of the observable but how far it is from commuting with the region's algebra; an observable supported far away sees nothing, however large it is. The commutator seminorm measures this, and we phrase hypothesis (G1) below with it.

Definition 1.2 (commutator seminorm)**.** For a region S and $c \in \mathcal{A}$,

$$N_S(c) := \sup\{\|[c, b]\| : b \in \mathcal{A}(S),\ \|b\| \le 1\}, \tag{1.1}$$

where $[c, b] = cb - bc$.

Proposition 1.3 (properties of the seminorm)**.** Lean ✓ *For regions $S \subseteq S'$, $c, c' \in \mathcal{A}$ and $z \in \mathbb{C}$:*

(a) $0 \le N_S(c) \le 2\|c\|$, *and* $\|[c, b]\| \le N_S(c)$ *for every* $b \in \mathcal{A}(S)$ *with* $\|b\| \le 1$*;*

(b) $N_S(c) = 0$ *when c commutes with every element of* $\mathcal{A}(S)$*;*

(c) $N_S(c) \le N_{S'}(c)$*;*

(d) $N_S(c + c') \le N_S(c) + N_S(c')$, $N_S(zc) = |z|\, N_S(c)$, *and* $|N_S(c) - N_S(c')| \le 2\|c - c'\|$.

Proof. The set in (1.1) contains 0 (take $b = 0$) and is bounded by $2\|c\|$, since $\|[c, b]\| \le 2\|c\|\|b\|$; this gives (a). In (b) every element of the set is 0. In (c) the set for S is contained in the set for S' because $\mathcal{A}(S) \subseteq \mathcal{A}(S')$. The first two parts of (d) follow from $[c + c', b] = [c, b] + [c', b]$ and $[zc, b] = z[c, b]$ by taking suprema; the last part from the first two applied to $c = (c - c') + c'$ and $c' = (c' - c) + c$ together with (a). □

1.3 Local generators

The dynamics of a lattice system is a one-parameter group or semigroup $A \mapsto e^{tL}A$ on the observables, and L is a sum of pieces, one per interaction term. For a Hamiltonian $H = \sum_i \Phi_i$ the pieces are the derivations $\mathrm{i}[\Phi_i, \cdot]$; for an open system they are local Lindblad generators. What the bound needs from a piece is not that it is a derivation but three properties: it acts on an observable only through the observable's failure to commute with its own patch; it spreads supports by at most its patch; and no sub-collection of pieces can amplify norms. Stating the hypotheses at this level is what makes Hamiltonian and open dynamics instances of one theorem.

Definition 1.4 (family of local generators)**.** A *family of local generators* on a net is a finite index set I and, for each $i \in I$, a finite region $S_i \subseteq \Lambda$ (the *support*), a bounded linear map $L_i : \mathcal{A} \to \mathcal{A}$ (the *generator*) and a number $\kappa_i \geq 0$ (the *strength*), such that
(G1) *locality in the argument:* $\|L_i c\| \leq \kappa_i \, N_{S_i}(c)$ for every $c \in \mathcal{A}$;
(G2) *support closure:* $L_i \, \mathcal{A}(Z) \subseteq \mathcal{A}(Z \cup S_i)$ for every region Z;
(G3) *contraction:* $\|e^{tL_J}\| \leq 1$ for every $J \subseteq I$ and every $t \geq 0$, where $L_J := \sum_{i \in J} L_i$.
The *full generator* is $L := L_I$, the *total strength* $K := \sum_{i \in I} \kappa_i$, and the *dynamics* is $A(t) := e^{tL}a$ for $a \in \mathcal{A}$ and $t \geq 0$. A set S *crosses* a region Y if $S \cap Y \neq \emptyset$ and $S \not\subseteq Y$, that is, S meets Y and its complement. For a region Y, $I^{\partial Y} := \{i \in I : S_i \text{ crosses } Y\}$ is the set of terms *crossing* Y.

The superscript in $I^{\partial Y}$ is for now a label for the terms crossing Y; the boundary ∂Y as a set of sites is defined in Chapter 3. The exponential is that of the Banach algebra of bounded operators on $\mathcal{A}$, $e^{tL} = \sum_{n \geq 0} t^n L^n / n!$. We ask for (G3) for every sub-family because the proof switches off the terms crossing the probe's region and evolves with the rest; the instances verify it uniformly (unitarity for a Hamiltonian family, Chapter 4; for a Lindblad family it is carried as a hypothesis, implied by complete positivity with unit preservation, Chapter 5), so nothing is lost by asking for all J at once, though the proof uses only the full family and the complements $I \setminus I^{\partial Y'}$ of the regions Y' that the recursion visits.

Proposition 1.5 (sub-families)**.** Lean ✓ *For a family of local generators, all $J, J' \subseteq I$, every region Y and all $a, c \in \mathcal{A}$:*
(a) $L_{J \cup J'} = L_J + L_{J'}$ *when* $J \cap J' = \emptyset$*; in particular* $L = L_{I^{\partial Y}} + L_{I \setminus I^{\partial Y}}$*;*
(b) $\|e^{tL}a\| \leq \|a\|$ *for* $t \geq 0$*;*
(c) $\|L_J c\| \leq \sum_{i \in J} \kappa_i \, N_{S_i}(c)$*.*

Proof. (a) The splitting of a finite sum over a disjoint union.
(b) (G3) for $J = I$.
(c) The triangle inequality and (G1) term by term. □

Example 1.6 (Hamiltonian terms)**.** Lean ✓ In a nonzero ambient algebra, $\mathcal{A} \neq 0$ (so that a unitary has norm one), let $\Phi_i \in \mathcal{A}(S_i)$ be self-adjoint, $L_i := \mathrm{i}[\Phi_i, \cdot]$ and $\kappa_i := \|\Phi_i\|$. Then (G1) holds because $\|L_i c\| = \|[c, \Phi_i]\| = \|\Phi_i\| \, \|[c, \Phi_i/\|\Phi_i\|]\| \leq \kappa_i N_{S_i}(c)$ when $\Phi_i \neq 0$, and trivially otherwise. (G2) holds because $\Phi_i c$ and $c\Phi_i$ lie in the algebra generated by $\mathcal{A}(Z)$ and $\mathcal{A}(S_i)$, which is contained in $\mathcal{A}(Z \cup S_i)$. (G3) holds because e^{tL_J} is conjugation by the unitary $e^{\mathrm{i}tH_J}$ with $H_J = \sum_{i \in J} \Phi_i$, an isometry.

This is the literature's setting: an interaction Φ with terms $\Phi(Z)$ on finite sets Z and the Heisenberg dynamics $\tau_t(A) = e^{\mathrm{i}tH} A e^{-\mathrm{i}tH}$.

1.4 Paths

A signal from the probe's region Y to the observable's region X must pass from term to term, the first crossing Y and each crossing the last. The combinatorial object is the sequence of terms it passes through, and the bound will count such sequences, each weighted by the product of the strengths of its terms. We fix two conventions here that matter later. Each term crosses the patch of the previous one, meeting it and its complement: a term inside the current patch cannot

carry the signal out of it, as Chapter 2 shows, so no term follows itself and no term is nested in its predecessor (a term may recur later, after another one). And a path ends at the first term that touches X, because the recursion stops there. Figure 2 shows the crossing condition; Example 1.9 and the chain of Chapter 2 show the first-contact condition.

Definition 1.7 (first-contact paths)**.** Let X, Y be regions and $k \geq 1$. A *path of length k from Y to X* is a sequence $(i_1, \dots, i_k) \in I^k$ such that

$$S_{i_1} \text{ crosses } Y, \qquad S_{i_{j+1}} \text{ crosses } S_{i_j} \ \ (1 \leq j < k), \qquad S_{i_j} \cap X = \emptyset \ \ (1 \leq j < k), \qquad S_{i_k} \cap X \neq \emptyset.$$

Its *weight* is $w(i_1, \dots, i_k) := \prod_{j=1}^{k} \kappa_{i_j}$. There is no path of length 0. The *path count* $P_k(Y, X)$ is the sum of the weights of the paths of length k from Y to X, and $P_0(Y, X) := 0$.

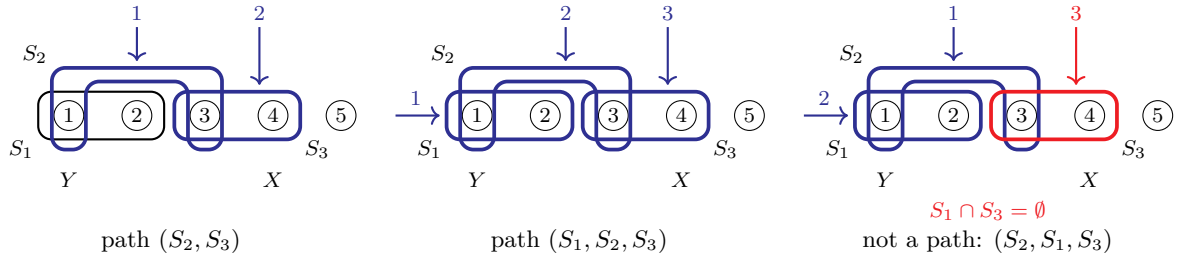


Figure 2: The interaction hypergraph of three terms $S_1 = \{1, 2\}$, $S_2 = \{1, 3\}$, $S_3 = \{3, 4\}$ on five sites, with $Y = \{1\}$ and $X = \{4\}$. Left: the path (S_2, S_3) of length two. Middle: the path (S_1, S_2, S_3) of length three; the sequence (S_2, S_2, S_3) is not a path, because S_2 does not cross itself. Right: the sequence (S_2, S_1, S_3) is not a path, because S_1 and S_3 do not meet, although each set meets the union of Y and the earlier ones. Blue marks the sets of the sequence in order, red the step that fails. For unit strengths the counts are $P_1 = 0$ and $P_k = 1$ for every $k \geq 2$, the single path being the alternating sequence $(S_2, S_1, S_2, \dots, S_3)$ or $(S_1, S_2, S_1, \dots, S_3)$ ending with S_2, S_3. With consecutive overlap in place of crossing, so that a term may follow itself, the first-contact counts would be $0, 1, 2, 4$; dropping first contact as well gives $0, 1, 3, 8$.

Proposition 1.8 (the path count by recursion)**.** Lean ✓ *For all regions X, Y:*

(a) $P_1(Y, X) = \sum_{i \in I^{\partial Y},\ S_i \cap X \neq \emptyset} \kappa_i$ *and, for* $k \geq 1$, $P_{k+1}(Y, X) = \sum_{i \in I^{\partial Y},\ S_i \cap X = \emptyset} \kappa_i \, P_k(S_i, X)$;

(b) $0 \leq P_k(Y, X) \leq K^k$ *for every k (with $K^0 = 1$).*

Proof. (a) The paths of length 1 are the single indices i whose support crosses Y and meets X, of weight κ_i. A sequence $(i_1, \dots, i_{k+1})$ is a path of length $k+1$ from Y to X exactly when S_{i_1} crosses Y and misses X, and $(i_2, \dots, i_{k+1})$ is a path of length k from S_{i_1} to X: the condition "S_{i_2} crosses S_{i_1}" is the first condition of the shorter path with Y replaced by S_{i_1}, and the remaining conditions are unchanged. Splitting the sum according to i_1 and using $w(i_1, \dots, i_{k+1}) = \kappa_{i_1} w(i_2, \dots, i_{k+1})$ gives the recursion.

(b) By induction on k using (a): $P_1 \leq \sum_i \kappa_i = K$ and $P_{k+1}(Y, X) \leq \sum_i \kappa_i K^k = K^{k+1}$. □

Example 1.9 (two qubits)**.** script ✓ Two sites $\Lambda = \{0, 1\}$, each carrying a qubit, so $\mathcal{A} = \mathrm{M}_2(\mathbb{C}) \otimes \mathrm{M}_2(\mathbb{C})$ with $\mathcal{A}(\{0\}) = \mathrm{M}_2(\mathbb{C}) \otimes 1$, $\mathcal{A}(\{1\}) = 1 \otimes \mathrm{M}_2(\mathbb{C})$, and the Pauli matrices $\sigma_j^x, \sigma_j^y, \sigma_j^z$ acting on site j. The Hamiltonian is the single term

$$H = \sigma_0^z \sigma_1^z,$$

so the family has one generator, $S_1 = \{0, 1\}$, $L_1 = \mathrm{i}[H, \cdot]$ and $\kappa_1 = \|H\| = 1$ (Example 1.6). Take the observable $a = \sigma_0^x$ on $X = \{0\}$ and the probe $b = \sigma_1^x$ on $Y = \{1\}$. The support S_1 crosses Y and meets X, so the only path is (1), of length one: $P_1 = 1$ and $P_k = 0$ for $k \geq 2$, since a path of length two or more would have to miss X at its first step.

The dynamics is explicit. Since H anticommutes with σ_0^x, $e^{\mathrm{i}tH}\sigma_0^x = \sigma_0^x e^{-\mathrm{i}tH}$, so $A(t) = \sigma_0^x e^{-2\mathrm{i}tH} = \sigma_0^x(\cos 2t - \mathrm{i}\sin 2t\,\sigma_0^z\sigma_1^z)$ and

$$[A(t), \sigma_1^x] = -\mathrm{i}\sin 2t\,\sigma_0^x\sigma_0^z[\sigma_1^z, \sigma_1^x] = 2\sin 2t\,\sigma_0^x\sigma_0^z\sigma_1^y,$$

of norm $2|\sin 2t|$. The path-sum bound of Chapter 2 will give $4t$.

Notes and references

The net of local algebras is the one of algebraic quantum field theory, restricted to a lattice. The bound goes back to [2]; its form for interactions on general lattices, decaying with the distance, is that of [3, 4]. The commutator seminorm is the quantity $\sup_{\|B\|=1}\|[A, B]\|$ that the review [1] uses for local approximants (Proposition 4.1 there), here promoted to the hypothesis (G1). The literature states the dynamics through an interaction Φ ([4], eqs. (2.5)–(2.6); [6], §2) or through edge Hamiltonians H_e ([1], §3); both are instances of Definition 1.4 (Chapters 4 and 5). The crossing condition is that of the chains of [6], eqs. (3.41)–(3.42), which run over sets meeting the previous set and its complement (their chains start at the observable's region and ours at the probe's; Chapter 2 reconciles the orientation); the sequences of [1], Theorem 3.4, eq. (3.34), run from a set meeting Y to a set meeting X with consecutive overlap and distinct consecutive terms, a larger class; the first-contact condition is ours. *Union-chains*, sequences in which each set meets the union of Y and the earlier sets, are a tempting but wrong bookkeeping object: the three-term example of Figure 2 separates the two notions, and there are four-term hypergraphs on three sites for which union-chains are not dominated by the site-matrix bound of Chapter 3, whereas paths always are.

Checks. The counts of Figure 2 and of Example 1.9 are computed by `checks/m1a_check.py` and `checks/path_check.py`; the four-term hypergraph is in the latter's section 2.

2 The path-sum bound

The commutator $[A(t), b]$ of an evolved observable with a probe b on a region Y can only be nonzero through the terms of the generator that cross the boundary of Y (a term inside Y cannot act on an observable supported outside it): with the crossing terms switched off, the observable's support never grows into Y, and causality kills the commutator. Duhamel's identity isolates exactly the contribution of the switched-off terms, as a time integral; hypothesis (G1) then replaces the probe b by a probe on the patch S_i of a single term, one step closer to the observable, and hypothesis (G3) guarantees that the free evolution in between cannot amplify anything.

Iterating, we bound the commutator by a sum over paths of terms leading from Y to X, each weighted by the product of its strengths and by the factor $t^k/k!$ that the nested time integrals produce; we call this sum the *path series*. This chapter proves that bound, Theorem 2.9, in five steps: the two analytic lemmas (Duhamel's identity and the invariance of closed subspaces under the exponential), the support of the restricted dynamics (a term inside the probe's region annihilates the observables outside it), the one-step inequality, the iteration with an explicit remainder, and the limit. Figure 3 shows the mechanism.

2.1 Two lemmas on the exponential

Lemma 2.1 (Duhamel's identity). Lean ✓ *Let $\mathbb{B}$ be a complete normed unital real algebra, $R, Q \in \mathbb{B}$ and $t \in \mathbb{R}$. Then*

$$e^{t(R+Q)} = e^{tR} + \int_0^t e^{(t-s)R}\, Q\, e^{s(R+Q)}\, ds, \tag{2.1}$$

the integrand being continuous in s.

Proof. Let $f(s) := e^{(t-s)R} e^{s(R+Q)}$. For every $G \in \mathbb{B}$ the map $u \mapsto e^{uG}$ is differentiable with derivative $Ge^{uG} = e^{uG}G$, so by the chain rule and the product rule

$$f'(s) = -e^{(t-s)R} R\, e^{s(R+Q)} + e^{(t-s)R}(R+Q)\, e^{s(R+Q)} = e^{(t-s)R}\, Q\, e^{s(R+Q)},$$

a continuous function of s. The fundamental theorem of calculus gives $f(t) - f(0) = \int_0^t f'(s)\, ds$, and $f(t) = e^{t(R+Q)}$, $f(0) = e^{tR}$. □

Lemma 2.2 (invariance). Lean ✓ *Let E be a complex Banach space, $M \subseteq E$ a closed complex subspace and R a bounded complex-linear operator with $R(M) \subseteq M$. Then $e^{tR}(M) \subseteq M$ for every $t \in \mathbb{R}$.*

Proof. For $m \in M$ the partial sums $\sum_{n \le N} t^n R^n m/n!$ lie in M, since $R^n m \in M$ by induction, and they converge in norm to $e^{tR}m$. M is closed. □

2.2 The restricted dynamics and the first term

Fix a causal net and a family of local generators. For a region Y, write $L^{\partial Y} := L_{I^{\partial Y}}$ for the terms crossing Y and $L' := L_{I \setminus I^{\partial Y}}$ for the rest, the terms disjoint from Y or inside Y, so that $L = L^{\partial Y} + L'$ (Proposition 1.5).

The terms inside Y belong with the free part: an observable supported outside Y commutes with everything on a patch inside Y, so its commutator seminorm on that patch vanishes, and (G1) then says the term does nothing to it. Such a term cannot carry a signal out of Y. Only the terms that straddle the boundary of Y can, and they are the perturbation.

Proposition 2.3 (a term inside the probe's region annihilates the complement). Lean ✓ *Let the net be causal, $S_i \subseteq Y$ and $c \in \mathcal{A}(Y^c)$. Then $L_i c = 0$.*

Proof. Every $b \in \mathcal{A}(S_i) \subseteq \mathcal{A}(Y)$ commutes with c, since Y^c and Y are disjoint, so $N_{S_i}(c) = 0$ by Proposition 1.3(b), and (G1) gives $\|L_i c\| \le \kappa_i \cdot 0$. □

Proposition 2.4 (support of the restricted dynamics). Lean ✓

(a) Let $J \subseteq I$ and let W be a region with $S_i \subseteq W$ for every $i \in J$. Then $L_J\, \mathcal{A}(W) \subseteq \mathcal{A}(W)$ and $e^{tL_J}\, \mathcal{A}(W) \subseteq \mathcal{A}(W)$ for every $t \in \mathbb{R}$.

(b) Let the net be causal. Then $L'\, \mathcal{A}(Y^c) \subseteq \mathcal{A}(Y^c)$ and $e^{tL'}\, \mathcal{A}(Y^c) \subseteq \mathcal{A}(Y^c)$ for every $t \in \mathbb{R}$.

Proof. (a) For $c \in \mathcal{A}(W)$ and $i \in J$, (G2) gives $L_i c \in \mathcal{A}(W \cup S_i) = \mathcal{A}(W)$, and a finite sum of elements of $\mathcal{A}(W)$ lies in it. Lemma 2.2 applies to the closed subspace $\mathcal{A}(W)$.

(b) A term $i \notin I^{\partial Y}$ either misses Y, so that $S_i \subseteq Y^c$ and (G2) keeps $\mathcal{A}(Y^c)$ as in (a), or lies inside Y, and then $L_i c = 0$ for $c \in \mathcal{A}(Y^c)$ by Proposition 2.3. Summing, L' maps $\mathcal{A}(Y^c)$ into itself, and Lemma 2.2 applies. □

Proposition 2.5 (the first term vanishes). Lean ✓ *Let the net be causal. If $X \cap Y = \emptyset$, $a \in \mathcal{A}(X)$ and $b \in \mathcal{A}(Y)$, then $[e^{tL'}a, b] = 0$ for every $t \in \mathbb{R}$.*

Proof. $X \subseteq Y^c$, so $a \in \mathcal{A}(Y^c)$ and $e^{tL'}a \in \mathcal{A}(Y^c)$ by Proposition 2.4(b); Y^c and Y are disjoint, so causality gives the claim. □

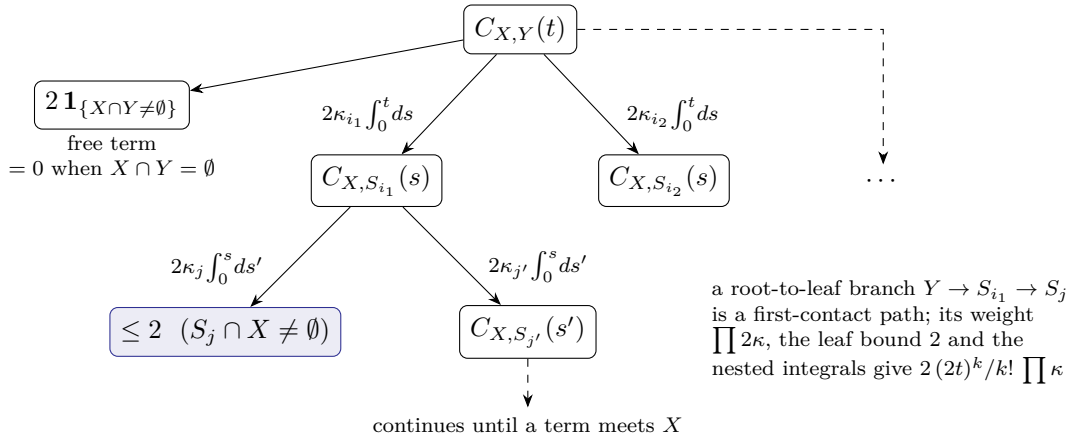


Figure 3: The probe-reset recursion. The commutator of the evolved observable $A(t)$ on X with a probe on Y is bounded, after one Duhamel step, by the terms S_i crossing Y, each contributing $2\kappa_i$ times a time integral of the same commutator with a probe on S_i. The recursion stops at the first term meeting X, where the trivial bound 2 is used; the chain of patches visited is a path of Definition 1.7.

2.3 The one-step inequality

Proposition 2.6 (the reset step). Lean ✓ *For $s \leq t$, $c \in \mathcal{A}$ and $b \in \mathcal{A}$ with $\|b\| \leq 1$,*

$$\|[e^{(t-s)L'}\, L^{\partial Y} c, b]\| \leq 2 \sum_{i \in I^{\partial Y}} \kappa_i\, N_{S_i}(c).$$

Proof. $\|[c', b]\| \leq 2\|c'\|$ for every $c' \in \mathcal{A}$ and $\|b\| \leq 1$; $\|e^{(t-s)L'}\| \leq 1$ by (G3) for the sub-family $I \setminus I^{\partial Y}$ and $t - s \geq 0$; and $\|L^{\partial Y} c\| \leq \sum_{i \in I^{\partial Y}} \kappa_i N_{S_i}(c)$ by Proposition 1.5(c). □

The reset step, the *probe reset*, is where the probe moves: on the right-hand side the commutator with the original probe b has become the seminorm $N_{S_i}(c)$, a supremum of commutators with probes on the single patch S_i. The next statement packages one application of Duhamel's identity with this step. It takes an arbitrary continuous majorant of the reset seminorms, because that is how the induction will consume it. The theorem needs the regions disjoint, so a probe on a patch that meets X cannot be reset again; there we use the trivial bound 2 of Proposition 1.3(a) instead. This is why paths end at first contact.

Theorem 2.7 (one Duhamel step). Lean ✓ *Let the net be causal, $X \cap Y = \emptyset$, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ with $\|b\| \leq 1$, $t \geq 0$, and let $\beta_i : \mathbb{R} \to \mathbb{R}$ $(i \in I^{\partial Y})$ be continuous functions with $N_{S_i}(A(s)) \leq \beta_i(s)$ for all $s \in [0, t]$. Then*

$$\|[A(t), b]\| \leq 2 \sum_{i \in I^{\partial Y}} \kappa_i \int_0^t \beta_i(s)\, ds. \tag{2.2}$$

Proof. Lemma 2.1 with $R = L'$ and $Q = L^{\partial Y}$, applied to a, gives $A(t) = e^{tL'}a + \int_0^t e^{(t-s)L'} L^{\partial Y} A(s)\, ds$. Evaluation at a and the commutator with b are bounded linear maps and pass through the integral:

$$[A(t), b] = [e^{tL'}a, b] + \int_0^t [e^{(t-s)L'} L^{\partial Y} A(s), b]\, ds.$$

The first term is 0 by Proposition 2.5. The norm of the integral is at most the integral of the norm, which Proposition 2.6 with $c = A(s)$ bounds pointwise by $2\sum_{i \in I^{\partial Y}} \kappa_i N_{S_i}(A(s)) \leq 2\sum_{i \in I^{\partial Y}} \kappa_i \beta_i(s)$, a continuous function. □

2.4 The iteration and the series

For $k \geq 0$ and $t \in \mathbb{R}$ put $\gamma_k(t) := (2t)^{k+1}/(k+1)!$, and for regions X, Y, $n \geq 0$ and $t \in \mathbb{R}$ (a polynomial in t, used for $t \geq 0$),

$$B_n(Y, X, t) := 2\Big(\sum_{k<n} \gamma_k(t)\, P_{k+1}(Y, X) + \gamma_n(t)\, K^{n+1}\Big), \tag{2.3}$$

the truncated series with its remainder. The identity $\int_0^t \gamma_k(s)\, ds = \gamma_{k+1}(t)/2$ is what makes the time-ordered integrals produce the factorials.

Theorem 2.8 (iteration). Lean ✓ *Let the net be causal, $X \cap Y = \emptyset$, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ with $\|a\|, \|b\| \leq 1$, and $t \geq 0$. Then $\|[A(t), b]\| \leq B_n(Y, X, t)$ for every $n \geq 0$.*

Proof. Induction on n, for all disjoint pairs X, Y, all $t \geq 0$ and all a, b of norm at most 1 at once.

Base. Theorem 2.7 with the constant majorants $\beta_i = 2$, legitimate since $N_{S_i}(A(s)) \leq 2\|A(s)\| \leq 2$ by Propositions 1.3(a) and 1.5(b), gives $\|[A(t), b]\| \leq 4t \sum_{i \in I^{\partial Y}} \kappa_i \leq 4tK = B_0(Y, X, t)$.

Step. Assume the bound for n. Apply Theorem 2.7 with the majorant $\beta_i(s) = 2$ when S_i meets X, and $\beta_i(s) = B_n(S_i, X, s)$ when S_i misses X; the latter is a majorant because, for every unit $b' \in \mathcal{A}(S_i)$, the induction hypothesis for the disjoint pair (X, S_i) gives $\|[A(s), b']\| \leq B_n(S_i, X, s)$, and $N_{S_i}(A(s))$ is the supremum of these. Integrating (2.3) with the identity for $\int \gamma_k$,

$$\int_0^t B_n(S_i, X, s)\, ds = \sum_{k<n} \gamma_{k+1}(t)\, P_{k+1}(S_i, X) + \gamma_{n+1}(t)\, K^{n+1}.$$

Summing over $i \in I^{\partial Y}$ with the weights $2\kappa_i$: the terms meeting X contribute $2\gamma_0(t) \sum \kappa_i = 2\gamma_0(t) P_1(Y, X)$; the terms missing X contribute $2 \sum_{k<n} \gamma_{k+1}(t) \sum_i \kappa_i P_{k+1}(S_i, X) = 2 \sum_{k<n} \gamma_{k+1}(t) P_{k+2}(Y, X)$ by the recursion of Proposition 1.8(a), and a remainder $2\gamma_{n+1}(t) K^{n+1} \sum_i \kappa_i \leq 2\gamma_{n+1}(t) K^{n+2}$, where the sum over the terms missing X is at most K. Together this is $B_{n+1}(Y, X, t)$: the first group is the $k = 0$ term of the sum in (2.3) for $n+1$, the second group its terms $k = 1, \ldots, n$, and the remainder its last term. □

Theorem 2.9 (the path-sum bound). Lean ✓ *Let the net be causal and $t \geq 0$. For regions X, Y, $a \in \mathcal{A}(X)$ and $b \in \mathcal{A}(Y)$,*

$$\|[A(t), b]\| \leq 2\|a\|\|b\| \Big(\mathbf{1}_{\{X \cap Y \neq \emptyset\}} + \sum_{k \geq 1} \frac{(2t)^k}{k!}\, P_k(Y, X)\Big), \tag{2.4}$$

where $\mathbf{1}_{\{P\}}$ is the indicator of a condition P, 1 if it holds and 0 otherwise; the series converges, with sum at most $e^{2tK} - 1$. Equivalently, for $C_{X,Y}(t) := \sup\{\|[e^{tL} a, b]\| : a \in \mathcal{A}(X), b \in \mathcal{A}(Y), \|a\|, \|b\| \leq 1\}$,

$$C_{X,Y}(t) \leq 2\, \mathbf{1}_{\{X \cap Y \neq \emptyset\}} + 2 \sum_{k \geq 1} \frac{(2t)^k}{k!}\, P_k(Y, X).$$

Proof. The k-th term is at most $(2tK)^k/k!$ by Proposition 1.8(b), so the series converges. If X meets Y the bound $2\|a\|\|b\|$ holds since $\|A(t)\| \leq \|a\|$. If $X \cap Y = \emptyset$ and a, b are unit, Theorem 2.8 gives, for every n, the truncated series plus $2(2tK)^{n+1}/(n+1)!$, which tends to 0; the truncated series is at most the full one. General a, b follow by scaling, and the supremum form by taking the supremum. □

Example 2.10 (two qubits, continued). script ✓ For $H = \sigma_0^z \sigma_1^z$, $a = \sigma_0^x$ and $b = \sigma_1^x$ (Example 1.9), $P_1 = 1$ and $P_k = 0$ for $k \geq 2$, so Theorem 2.9 gives $\|[A(t), b]\| \leq 2 \cdot 2t = 4t$. The exact value $2|\sin 2t|$ lies below $4t$ and matches it to first order: the constant 2κ per step is not loose at short times.

Example 2.11 (a three-qubit chain). script ✓ Three qubits on $\Lambda = \{0, 1, 2\}$ with the Hamiltonian

$$H = \sigma_0^z \sigma_1^z + \tfrac{1}{2}\,\sigma_1^x \sigma_2^x,$$

that is, two terms: $\Phi_1 = \sigma_0^z \sigma_1^z$ on $S_1 = \{0, 1\}$ with $\kappa_1 = 1$, and $\Phi_2 = \frac{1}{2}\sigma_1^x \sigma_2^x$ on $S_2 = \{1, 2\}$ with $\kappa_2 = \frac{1}{2}$. The observable is $a = \sigma_0^x$ on $X = \{0\}$, the probe $b = \sigma_2^z$ on $Y = \{2\}$.

The paths. Only S_2 crosses Y, and it misses X; after it only S_1 can follow, since S_1 crosses S_2 while S_2 does not cross itself, and S_1 meets X. So the only path is (S_2, S_1): $P_2 = \kappa_1 \kappa_2 = \frac{1}{2}$, every other P_k is 0, and Theorem 2.9 gives

$$\|[A(t), \sigma_2^z]\| \le 2 \cdot \frac{(2t)^2}{2!} \cdot \frac{1}{2} = 2t^2 \qquad \text{for every } t \ge 0.$$

The exact value. The three Pauli strings $\mathbf{e}_0 = \sigma_0^x$, $\mathbf{e}_1 = \sigma_0^y \sigma_1^z$, $\mathbf{e}_2 = \sigma_0^y \sigma_1^y \sigma_2^x$ satisfy $[H, \mathbf{e}_0] = 2\mathrm{i}\mathbf{e}_1$, $[H, \mathbf{e}_1] = -2\mathrm{i}\mathbf{e}_0 - \mathrm{i}\mathbf{e}_2$, $[H, \mathbf{e}_2] = \mathrm{i}\mathbf{e}_1$, so $A(t) = f_0(t)\mathbf{e}_0 + f_1(t)\mathbf{e}_1 + f_2(t)\mathbf{e}_2$ with $f_0' = 2f_1$, $f_1' = -2f_0 - f_2$, $f_2' = f_1$ and $(f_0, f_1, f_2)(0) = (1, 0, 0)$; hence $f_2(t) = \frac{2}{5}(\cos\sqrt{5}\,t - 1)$. Only $\mathbf{e}_2$ fails to commute with the probe, $[\mathbf{e}_2, \sigma_2^z] = -2\mathrm{i}\,\sigma_0^y \sigma_1^y \sigma_2^y$, so

$$\|[A(t), \sigma_2^z]\| = 2|f_2(t)| = \tfrac{4}{5}(1 - \cos\sqrt{5}\,t) = 2t^2 - \tfrac{5}{6}t^4 + O(t^6).$$

The ratio to the bound is $1 - \frac{5}{12}t^2 + \dots$, equal to 0.99996 at $t = 0.01$ and decreasing; the bound is never exceeded. On this chain the path-sum bound is sharp to leading order at short times, while at long times every commutator is at most 2 and the bound grows.

Route against the source

The sources differentiate the commutator in time and move the derivative onto the probe with the Jacobi identity. Writing $f(t) = [\tau_t(A), B]$ and $\frac{d}{dt}\tau_t(A) = \mathrm{i}[\tau_t(H_X), \tau_t(A)]$ with H_X the terms meeting the observable's region, [6] (eqs. (3.40)–(3.48)) and [4] obtain $f' = \mathrm{i}[\tau_t(H_X), f] - \mathrm{i}[\tau_t(A), [\tau_t(H_X), B]]$; the first term generates a norm-preserving flow and drops out of the norm estimate, the second resets the observable to a single term $\Phi(Z)$ with $Z \cap X \neq \emptyset$, and iterating gives a series over chains of overlapping supports. The review [1] (eqs. (3.26)–(3.29)) does the same in graph language.

The route above differs in three respects. It integrates (Duhamel) instead of differentiating, so that no Jacobi identity and no derivation property are needed, and the hypotheses reduce to (G1)–(G3); the open-system bound of Chapter 5 is therefore the same theorem. It resets the probe rather than the observable, which is a matter of taste. And it stops at first contact, which the sources do not. The path class is otherwise theirs.

The chains of [6], eqs. (3.41)–(3.42), are sequences $Z_1, \dots, Z_n$ with Z_1 meeting the observable's region and its complement, each Z_{j+1} meeting Z_j and its complement, and Z_n meeting the probe's region; they run from the observable to the probe because the source resets the observable. For an automorphism group $\|[\tau_t(A), B]\| = \|[A, \tau_{-t}(B)]\|$ and the source's bound depends on $|t|$, so its theorem applied to the pair (B, A) runs the chains from Y: Z_1 crosses Y and each Z_{j+1} crosses Z_j, which is Definition 1.7 without first contact. With one term per support and $\kappa_Z = \|\Phi(Z)\|$, the weights $2\|\Phi(Z_j)\|$ and the nested integrals $t^n/n!$ are the same, so in the Hamiltonian case the series (2.4) is at most the chain series of [6] term by term.

The reason the sources may restrict to crossing sets is the one of Proposition 2.3: a term inside the current region cannot move an observable supported outside it. The review's recursion (3.29b) drops the repeated edge without giving this reason; Proposition 2.3 supplies it.

The review's Theorem 3.4 sums over sequences from a set meeting Y to a set meeting X with consecutive overlap and distinct consecutive terms, a class containing ours since a set crossing another differs from it, so (2.4) is at most that series as well; the review attributes the theorem to the work it numbers [34], and the stronger irreducible-path bound of [8], Theorem 3, is not claimed here.

Notes and references

Theorem 2.9 is the generator-level form of the Lieb-Robinson path series, Theorem 3.4 of [1] and eqs. (3.41)–(3.42) of [6], with the differences listed above; its constants, $2\kappa_i$ per step and $t^k/k!$, are theirs. Duhamel's identity and the invariance lemma are standard.

Checks. The numbers of Example 2.11 are computed by `checks/m1a_crossing_check.py` and `checks/m1a_chain_expansion_check.py`, the latter also for the closed form of the commutator norm.

3 Closed forms and light cones

The path-sum bound of Chapter 2 is exact bookkeeping: it counts the sequences of terms that can carry a signal. To read it as a statement about distances, we replace the count of paths by something that only sees pairs of sites, the *site matrix*, whose entry at (u, u') adds up the strengths of the terms containing both sites. A path of k terms visits $k+1$ sites, one in the probe's region, one in each of the $k-1$ overlaps of consecutive supports, and one in the observable's region; its weight is at most the product of the matrix entries along those sites. Summing over all site sequences gives an entry of the k-th power of the matrix, and summing the series gives its exponential.

From there the published forms follow by three estimates: a decay function bounds the entries, a bounded geometry bounds the row sums and makes distant entries of the powers vanish, and the tail of the exponential series gives the light cone. A last section extends all of these to negative times when the negated generators are contractive as well, as they are for a Hamiltonian.

3.1 The site matrix and walks

Throughout, $(S_i, L_i, \kappa_i)_{i\in I}$ is a family of local generators on a causal net over an arbitrary set of sites Λ, and $T := \bigcup_i S_i$ is the finite set of *touched sites*, the sites the family touches. We write $\chi_i(u) := \mathbf{1}_{\{u\in S_i\}}$ for the indicator of a support.

Definition 3.1 (site matrix, walks)**.** The *site matrix* of the family is

$$h_{uu'} := \sum_{i\in I:\ u\in S_i,\ u'\in S_i} \kappa_i = \sum_i \kappa_i\, \chi_i(u)\, \chi_i(u') \qquad (u, u' \in \Lambda), \tag{3.1}$$

so that $h_{uu} = \sum_{i\ni u} \kappa_i$. The *walk weights* are defined by $w_0(y, x) := \mathbf{1}_{\{y=x\}}$ and

$$w_{k+1}(y, x) := \sum_{z\in T} h_{yz}\, w_k(z, x). \tag{3.2}$$

For a region Y, the *boundary* $\partial Y := Y \cap \bigcup_{i\in I^{\partial Y}} S_i$ is the set of sites of Y touched by a term crossing Y. For regions Y, X put

$$W_k(Y, X) := \sum_{y\in\partial Y} \sum_{x\in X\cap T} w_k(y, x). \tag{3.3}$$

$w_k(y, x)$ is the total weight of the walks of k steps from y to x through the touched sites, each step $z \to z'$ weighted by $h_{zz'}$; on a finite lattice it is the entry $(h^k)_{yx}$ (Proposition 3.5). A walk may stay at a site, weighted by the diagonal of h, or return to one. The boundary ∂Y is the part of the probe's region that the dynamics can reach from outside: a term inside Y does nothing to an observable supported outside Y (Proposition 2.3), so a signal enters Y only through the sites that a crossing term touches. Both ∂Y and $X \cap T$ are finite, so W_k is a finite sum.

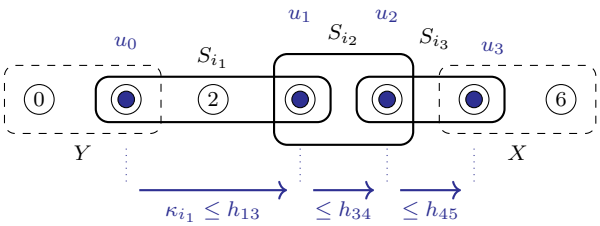


Figure 4: A path of three terms from the probe's region Y to the observable's region X, and a sequence of sites along it: $u_0 \in S_{i_1} \cap Y$ lies on the boundary ∂Y, each u_j in the overlap $S_{i_j} \cap S_{i_{j+1}}$, and $u_3 \in S_{i_3} \cap X$. Each step $u_{j-1} \to u_j$ stays inside the support S_{i_j}, so $\kappa_{i_j} \le h_{u_{j-1}u_j}$: the weight of the path is at most the weight of this walk. Theorem 3.3 makes the idea precise without choosing the sites, by expanding the walk sum through the indicators of the supports.

Proposition 3.2 (properties). Lean ✓ *Let* $K_1 := \sum_i \kappa_i |S_i|$.

(a) $h_{uu'} \ge 0$, $h_{uu'} = h_{u'u}$, *and* $h_{uu'} = 0$ *unless* $u, u' \in T$;

(b) $\sum_{u' \in T} h_{uu'} \le K_1$ *for every site* u;

(c) $w_k(y, x) \ge 0$; *for* $k \ge 1$, $w_k(y, x) = 0$ *unless* $y, x \in T$, *and* $w_1(y, x) = h_{yx}$;

(d) $\sum_{x \in T} w_k(y, x) \le q^k$ *whenever every row sum* $\sum_{u' \in T} h_{uu'}$ *is at most* $q \ge 0$; *hence* $W_k(Y, X) \le |\partial Y|\, q^k$, *and in particular* $W_k(Y, X) \le |\partial Y|\, K_1^k$.

Proof. (a) Read off the definitions.

(b) Exchanging the sums, $\sum_{u' \in T} \sum_{i:\, u, u' \in S_i} \kappa_i = \sum_{i \ni u} \kappa_i |S_i| \le K_1$.

(c) Read off the definitions.

(d) By induction on k: for $k = 0$ the sum is $\mathbf{1}_{\{y \in T\}} \le 1$, and $\sum_x w_{k+1}(y, x) = \sum_z h_{yz} \sum_x w_k(z, x) \le q^k \sum_z h_{yz} \le q^{k+1}$. □

The proof of (b) gives the row sum exactly: $\sum_{u' \in T} h_{uu'} = \sum_{i \ni u} \kappa_i |S_i|$, the total weighted size of the supports that contain u.

Theorem 3.3 (domination of the path count). Lean ✓ *For every* $k \ge 1$ *and all regions* X, Y,

$$P_k(Y, X) \le W_k(Y, X). \tag{3.4}$$

Proof. Induction on k, the claim quantified over all Y.

$k = 1$. $P_1(Y, X)$ is the sum of κ_i over the terms crossing Y and meeting X. Expanding (3.3) with $w_1 = h$ and (3.1),

$$W_1(Y, X) = \sum_i \kappa_i \Big(\sum_{y \in \partial Y} \chi_i(y) \Big) \Big(\sum_{x \in X \cap T} \chi_i(x) \Big).$$

For a term i crossing Y and meeting X, some site of S_i lies in Y, hence in ∂Y, and some site of S_i lies in X, hence in $X \cap T$; so both brackets are at least 1. Every other term contributes a nonnegative amount. Hence $P_1 \le W_1$.

Step, the left side. The recursion of Proposition 1.8 gives $P_{k+1}(Y, X) = \sum_{i \in Q} \kappa_i P_k(S_i, X)$, with Q the terms crossing Y and missing X. We extend the notation: $w_k(z, X) := \sum_{x \in X \cap T} w_k(z, x) \ge 0$. By the induction hypothesis for the region S_i, and since $\partial S_i \subseteq S_i$, $P_k(S_i, X) \le W_k(S_i, X) \le \sum_{z \in S_i} w_k(z, X)$.

Step, the right side. Expanding (3.3) with (3.2) and (3.1), and summing over z first,

$$W_{k+1}(Y, X) = \sum_i \kappa_i \Big(\sum_{y \in \partial Y} \chi_i(y) \Big) \sum_{z \in S_i} w_k(z, X).$$

Comparison. For $i \in Q$ the bracket is at least 1, as in the base case, and every other term is nonnegative. So $P_{k+1}(Y, X) \le \sum_{i \in Q} \kappa_i \sum_{z \in S_i} w_k(z, X) \le W_{k+1}(Y, X)$. □

What the domination loses: a walk may return to a site it has visited, while a path never repeats a term immediately, and a walk continues after reaching X, while a path stops. Both effects are visible in Example 3.6, where equality holds at $k = 2$ and the walk weights are strictly larger from $k = 3$ on.

3.2 The walk series and the matrix exponential

Theorem 3.4 (closed form (i): the walk series). Lean ✓ *Let the net be causal, $t \geq 0$, $a \in \mathcal{A}(X)$ and $b \in \mathcal{A}(Y)$. Then*

$$\|[A(t), b]\| \leq 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \sum_{k\geq 1} \frac{(2t)^k}{k!}\, W_k(Y, X)\Big), \tag{3.5}$$

the series converging with sum at most $|\partial Y|\,(e^{2K_1 t} - 1)$.

Proof. Termwise, $(2t)^k P_k/k! \leq (2t)^k W_k/k! \leq |\partial Y|(2tK_1)^k/k!$ by Theorem 3.3 and Proposition 3.2(d); so the walk series converges by comparison with the exponential series, and (2.4) gives (3.5). □

The terms of this series are the *walk terms*: for the pair (Y, X) and $k \geq 0$,

$$\omega_k(t) := \frac{(2t)^{k+1}}{(k+1)!}\, W_{k+1}(Y, X) = \gamma_k(t)\, W_{k+1}(Y, X), \tag{3.6}$$

with the coefficient γ_k of (2.3), so that the series in (3.5) is $\sum_{k\geq 0} \omega_k(t)$. We shift the index by one so that ω_k carries the walks of $k+1$ steps as γ_k carries the paths of length $k+1$. The walk terms depend on the strengths κ_i, through the site matrix in W_{k+1}, and on the pair (Y, X); Chapter 5 uses them.

Proposition 3.5 (finite lattice). Lean ✓ *Let Λ be finite and regard h as a matrix. Then $w_k(y, x) = (h^k)_{yx}$, and for $s \in \mathbb{R}$ the entries of the matrix exponential are the series of the entries of the powers, $(e^{sh})_{yx} = \sum_{n\geq 0} s^n (h^n)_{yx}/n!$. Consequently*

$$\|[A(t), b]\| \leq 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \sum_{y\in\partial Y}\sum_{x\in X\cap T} \big(e^{2th} - 1\big)_{yx}\Big). \tag{3.7}$$

Proof. $w_{k+1}(y, x) = \sum_{z\in T} h_{yz}(h^k)_{zx} = \sum_{z\in\Lambda} h_{yz}(h^k)_{zx}$, the terms with $z \notin T$ vanishing, which is $(h^{k+1})_{yx}$. The exponential series converges in any algebra norm on the matrices, and every entry is bounded by the norm, so the entries converge to the entries. Then (3.7) is (3.5) with the series summed entrywise, over the pairs of $\partial Y \times (X \cap T)$ as in (3.3). □

The inner sum in (3.7) may equally run over all of X: for $x \notin T$ the column x of h vanishes, so $(h^n)_{yx} = 0$ for $n \geq 1$ and the entry $(e^{2th} - 1)_{yx}$ is zero.

Inequality (3.7) is the matrix form of the Lieb-Robinson bound in [1], with the sum over the probe's region sharpened to its boundary.

Example 3.6 (the three-qubit chain, continued). paper For $H = \sigma_0^z\sigma_1^z + \frac{1}{2}\sigma_1^x\sigma_2^x$ of Example 2.11, with $S_1 = \{0, 1\}$, $\kappa_1 = 1$, $S_2 = \{1, 2\}$, $\kappa_2 = \frac{1}{2}$, $X = \{0\}$ and $Y = \{2\}$: $T = \{0, 1, 2\}$, $\partial Y = \{2\}$, and

$$h = \begin{pmatrix} 1 & 1 & 0 \\ 1 & \frac{3}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The walk weights from the boundary site 2 to the observable's site 0 follow from (3.2): $w_1(2, 0) = h_{20} = 0$; $w_2(2, 0) = h_{21}w_1(1, 0) + h_{22}w_1(2, 0) = \frac{1}{2}$; and, with $w_2(1, 0) = h_{10}w_1(0, 0) + h_{11}w_1(1, 0) + h_{12}w_1(2, 0) = 1 + \frac{3}{2} = \frac{5}{2}$, $w_3(2, 0) = h_{21}w_2(1, 0) + h_{22}w_2(2, 0) = \frac{5}{4} + \frac{1}{4} = \frac{3}{2}$. The path counts are $P_2 = \frac{1}{2}$ and $P_k = 0$ otherwise: equality at $k = 2$, and from $k = 3$ on the walks revisit sites while the paths have stopped. The path series is exactly $2t^2$; the walk series (3.5) is $2\big(\frac{(2t)^2}{2!}\cdot\frac{1}{2} + \frac{(2t)^3}{3!}\cdot\frac{3}{2} + \dots\big) = 2t^2 + 4t^3 + \cdots = 2\big(e^{2th} - 1\big)_{20}$.

3.3 Decay functions

A decay function is a function of pairs of sites that reproduces itself under convolution; the bounds use a constant multiple of it that dominates the site matrix.

Definition 3.7 (decay function)**.** A *decay function* is a function $F : \Lambda \times \Lambda \to [0, \infty)$ with a constant $C_F > 0$ such that for all x, y and every finite $T' \subseteq \Lambda$,

$$\sum_{z \in T'} F(x, z)\, F(z, y) \le C_F\, F(x, y). \tag{3.8}$$

An *F-bound* of the family is a number $N_F \ge 0$ with $h_{xy} \le N_F F(x, y)$ for all x, y. Write $\Sigma_F(Y, X) := \sum_{y \in \partial Y} \sum_{x \in X \cap T} F(y, x)$.

Example 3.8 (a decay function on the chain)**.** paper On a finite chain $\Lambda = \{0, \dots, n\}$ with $d(x, y) = |x - y|$ and nearest-neighbour terms (the constants below do not depend on n), the function $F(x, y) = e^{-\lambda|x-y|}$ with $\lambda > 0$ is not enough: $\sum_z F(x, z)F(z, y)$ counts the $|x - y| + 1$ sites between x and y with the full weight $e^{-\lambda|x-y|}$, so no constant C_F works uniformly in n.

The standard remedy is a polynomial factor, $F(x, y) = e^{-\lambda|x-y|}/(1+|x-y|)^2$: the exponentials obey $e^{-\lambda(|x-z|+|z-y|)} \le e^{-\lambda|x-y|}$, and with $m = y - x$ and $j = z - x$, $(1 + |j|)(1 + |m - j|) \ge \frac{1}{2}(1+|m|)(1+\min\{|j|, |m-j|\})$, so $\sum_z F(x, z)F(z, y) \le C_F\, F(x, y)$ with $C_F = \sup_m \sum_{j \in \mathbb{Z}} 4/(1+\min\{|j|, |m - j|\})^2 < \infty$, uniformly in m (the constant can be sharpened). For the nearest-neighbour family with all $\kappa_i = 1$, the least F-bound is $N_F = 4e^{\lambda}$: the diagonal needs $h_{xx} = 2 \le N_F$ and a neighbouring pair needs $h_{x,x+1} = 1 \le N_F e^{-\lambda}/4$.

In general, for a distance d (Definition 3.10) and a non-increasing $F_0 : [0, \infty) \to (0, \infty)$, the function $F(x, y) = F_0(d(x, y))$ satisfies (3.8) as soon as the reproducing constant $C_F = \sup_{x,y} \sum_z F_0(d(x, z))F_0(d(z, y))/F_0(d(x, y))$ is finite, which is an assumption on F_0 and the lattice; for the lattice $\mathbb{Z}^\nu$ Chapter 4 quotes the constant from the literature and computes it on finite pieces. The least F-bound is then the literature's F-norm of the interaction, when each support carries one term (Chapter 4); every larger number is an F-bound as well.

Theorem 3.9 (closed form (ii): decay functions)**.** Lean ✓ *Under the hypotheses of Theorem 3.4, for a decay function F with constant C_F and an F-bound N_F,*

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X \cap Y \ne \emptyset\}} + \frac{e^{2N_F C_F t} - 1}{C_F}\, \Sigma_F(Y, X)\Big). \tag{3.9}$$

Proof. By induction on k, $w_k(y, x) \le N_F^k C_F^{k-1} F(y, x)$ for $k \ge 1$: the case $k = 1$ is $h \le N_F F$, and the step is $w_{k+1}(y, x) = \sum_{z \in T} h_{yz} w_k(z, x) \le N_F^{k+1} C_F^{k-1} \sum_{z \in T} F(y, z)F(z, x) \le N_F^{k+1} C_F^k F(y, x)$ by (3.8). Summing over ∂Y and $X \cap T$, $W_k(Y, X) \le N_F^k C_F^{k-1} \Sigma_F(Y, X)$, and the series in (3.5) is at most $(\Sigma_F/C_F) \sum_{k \ge 1} (2N_F C_F t)^k/k!$. □

With $F(x, y) = F_0(d(x, y))$, inequality (3.9) is the decay-function form of [6]. Their form offers a second option, the sum over the boundary of the observable's region instead of the probe's: it is the same statement with the roles of the two observables exchanged, which for a group of automorphisms follows from the identity $\|[\tau_t(A), B]\| = \|[A, \tau_{-t}(B)]\|$ of Theorem 4.6 (Chapter 4), with Theorem 3.16 for the negative time.

3.4 Bounded geometry and light cones

Definition 3.10 (bounded geometry)**.** Let $d : \Lambda \times \Lambda \to \mathbb{R}$ satisfy $d(u, u) = 0$ and $d(u, u'') \le d(u, u') + d(u', u'')$. The family has *bounded geometry* (g, ℓ, p, ρ) if every site lies in the supports of at most g indices i, every $\kappa_i \le \ell$, every $|S_i| \le p$, and $d(u, u') \le \rho$ for u, u' in a common support $(\ell, \rho \ge 0)$. Write $v := 2g\ell p$.

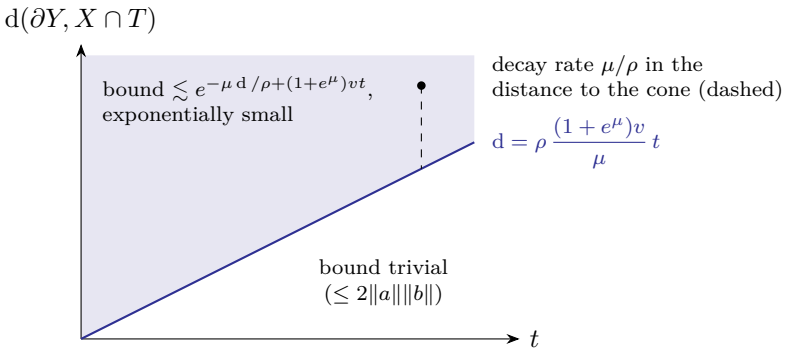


Figure 5: The light cone of Theorem 3.14, for $\mu, \rho > 0$: in the plane of the distance between the probe's boundary and the observable's region against the time t, the bound is at most a constant times $e^{-\mu\,\mathrm{d}/\rho+(1+e^{\mu})vt}$, so it is small outside the cone $\mathrm{d} > \rho(1+e^{\mu})vt/\mu$ (shaded) and trivial inside it. The slope of the line is schematic. The velocity $(1+e^{\mu})v\rho/\mu$ and the decay rate μ/ρ are those of the bound, not of the dynamics.

We call a function d with $d(u,u) = 0$ and the triangle inequality a *distance*. It is not assumed symmetric, so $d(u,u')$ and $d(u',u)$ may differ, and we keep the order of the arguments throughout: the first on the probe's side, the second on the observable's.

Proposition 3.11 (walks and distance). Lean ✓ *With bounded geometry (g,ℓ,p,ρ):*
(a) $w_k(y,x) = 0$ whenever $k\rho < d(y,x)$;
(b) every row sum of h is at most $g\ell p$, so $W_k(Y,X) \le |\partial Y|\,(g\ell p)^k$.

Proof. (a) By induction: a nonzero term $h_{yz}w_k(z,x)$ of (3.2) has y,z in a common support, so $d(y,z) \le \rho$, and $d(z,x) \le k\rho$; the triangle inequality gives $d(y,x) \le (k+1)\rho$.
(b) The row sum at u is $\sum_{i\ni u}\kappa_i|S_i|$ (the paragraph after Proposition 3.2), a sum of at most g terms, each at most ℓp. □

Lemma 3.12 (tail of the exponential series). Lean ✓ *For $x \ge 0$ and $r \in \mathbb{N}$: $\sum_{k\ge r} x^k/k! \le (x^r/r!)\,e^x$, since $x^{r+j}/(r+j)! \le (x^r/r!)(x^j/j!)$ by $r!\,j! \le (r+j)!$.*

Theorem 3.13 (closed form (iii): bounded geometry). Lean ✓ *Under the hypotheses of Theorem 3.4 and bounded geometry (g,ℓ,p,ρ), let $r \ge 1$ satisfy $(r-1)\rho < d(y,x)$ for all $y \in \partial Y$, $x \in X \cap T$. Then*

$$\|[A(t),b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + |\partial Y|\,\frac{(vt)^r}{r!}\,e^{vt}\Big), \qquad v = 2g\ell p. \tag{3.10}$$

Proof. For $k < r$ every pair in (3.3) is farther than $k\rho$, so $W_k(Y,X) = 0$ by Proposition 3.11(a); for $k \ge r$, $W_k \le |\partial Y|(g\ell p)^k$ by (b). The series in (3.5) is therefore at most $|\partial Y|\sum_{k\ge r}(vt)^k/k!$, and Lemma 3.12 applies. □

Theorem 3.14 (closed form (iv): the exponential light cone). Lean ✓ *Under the hypotheses of Theorem 3.13, for every real μ,*

$$\|[A(t),b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + |\partial Y|\,e^{vt}\,e^{-\mu r}(e^{e^{\mu}vt} - 1)\Big). \tag{3.11}$$

Proof. For $r \ge 1$ and $x \ge 0$, $x^r/r! = e^{-\mu r}(e^{\mu}x)^r/r! \le e^{-\mu r}\sum_{j\ge 1}(e^{\mu}x)^j/j! = e^{-\mu r}(e^{e^{\mu}x} - 1)$; insert $x = vt$ into (3.10). □

Take $\mu, \rho > 0$, regions with ∂Y and $X \cap T$ nonempty, and let $\mathrm{d}(\partial Y, X\cap T) := \min\{d(y,x) : y \in \partial Y,\ x \in X\cap T\}$ be positive; put $r = \lceil \mathrm{d}(\partial Y, X\cap T)/\rho \rceil \ge 1$. The hypothesis of Theorem 3.13 holds, and (3.11) is at most a constant times $e^{-\mu\,\mathrm{d}(\partial Y, X\cap T)/\rho+(1+e^{\mu})vt}$: a light cone of velocity $(1+e^{\mu})v\rho/\mu$ with decay rate μ/ρ outside it (Figure 5).

Example 3.15 (a nearest-neighbour chain)**.** paper Nearest-neighbour terms on the chain $\{0,\dots,n\}$, one term $\{j,j+1\}$ per bond with $\kappa_j \le \ell$: each site lies in at most $g=2$ supports of size $p=2$ and diameter $\rho=1$, so $v=8\ell$. For an observable at the site 0 and a probe at the site $n\ge 1$, the boundary is $\partial Y=\{n\}$ and $r=n$, and Theorem 3.13 reads $\|[A(t),b]\| \le 2\|a\|\|b\|\,(8\ell t)^n e^{8\ell t}/n!$. With $\mu=1$, Theorem 3.14 gives a light cone of velocity $(1+e)\cdot 8\ell \approx 30\ell$ and decay rate 1 per site outside it. The velocity is an artefact of the walk bound; on the three-site chain of Example 2.11 the path series itself is sharp to leading order at short times.

3.5 Time reversal

We proved the path-sum bound for $t\ge 0$, because hypothesis (G3) is a statement about e^{tL_J} for $t\ge 0$. If the family also satisfies (G3) with every L_i replaced by $-L_i$, which is the case for a group of automorphisms, we say that the family has a *contractive negation*, and then the bound holds for all times.

Theorem 3.16 (time reversal)**.** Lean ✓ *Let the net be causal, and suppose* $\|e^{-tL_J}\| \le 1$ *for every* $J\subseteq I$ *and* $t\ge 0$*. Then* (2.4)*,* (3.5) *and* (3.9) *hold for every* $t\in\mathbb{R}$ *with* $|t|$ *in place of* t*.*

Proof. The negated family $(S_i,-L_i,\kappa_i)$ satisfies (G1) and (G2), since $\|-L_ic\|=\|L_ic\|$ and $\mathcal{A}(Z\cup S_i)$ is a subspace, and (G3) by hypothesis; it has the same supports and strengths, hence the same paths, counts, site matrix and walk weights, and its dynamics at time $|t|$ is $e^{|t|(-L)}=e^{tL}$ for $t<0$. Apply the bounds to it. □

The matrix form (3.7) and the bounded-geometry and light-cone forms (3.10) and (3.11) follow from (3.5) alone, by the proofs of Proposition 3.5 and Theorems 3.13 and 3.14. Applied to the negated family, which has the same supports and strengths, they too hold for every $t\in\mathbb{R}$ with $|t|$ in place of t under the hypothesis of Theorem 3.16.

Notes and references

Site matrix. The review [1] defines $h_{uu'}$ for two-site terms, with $\|H_{uu'}\|$ off the diagonal and the sum of the strengths at a site on it (its eq. (3.35)); [8] writes $h_{ij}=\sum_{X\ni i,j}\|H_X\|$ for general supports. Definition 3.1 is the latter with κ_i in place of $\|H_X\|$, and (3.7) is the review's (3.36), with its constant doubled and the -1 that the series proof gives; the review's commutator quantity is half of $C_{X,Y}(t)$.

Boundary and decay functions. The boundary ∂Y is the Φ-boundary of [6], eq. (3.20). Decay functions F_0 with a reproducing constant and the F-norm are [6], eqs. (3.8)–(3.9) and (3.14); Theorem 3.1 of [6] is (3.9) with $F=F_0\circ d$ and N_F the F-norm, the exponent $2N_FC_Ft$ being their $2I_{t,s}$ for a time-independent interaction. Only the finite-sum form (3.8) enters here, so neither a metric nor the convergence of $\sum_y F_0(d(x,y))$ is needed before Chapter 4.

Geometry. The bounded-degree form is the review's Proposition 3.5, whose constants $(4(g-1)h$ for the velocity and $K=1/(1-2/e)$ for the tail) are sharper than (3.10)'s $v=2g\ell p$ and e^{vt}, because the review's walks never return along the edge they came from; the exponential form is the review's Example 3.6.

Checks. The site matrix, walk weights and series of Example 3.6, the domination on random hypergraphs, the constants C_F (the least constant on a chain of 40 sites at $\lambda=\frac12$ is 3.72) and $N_F=4e^{\lambda}$ of Example 3.8, the vanishing of $(h^k)_{xy}$ below the distance, the tail bound behind Example 3.15 and the comparison with the review's Proposition 3.5 are computed by `checks/m1b_check.py`.

4 Hamiltonians and qudit lattices

We stated the theorem of Chapter 2 for bounded generators with three properties, and Example 1.6 sketched why Hamiltonian terms have them. This chapter proves it once for all. A Hamiltonian built from self-adjoint local terms Φ_i generates the derivations $a \mapsto \mathrm{i}[\Phi_i, a]$, and the three properties are three facts about a C*-algebra: locality in the argument is the homogeneity of the commutator seminorm, support closure is isotony, and contraction, in both time directions, is the unitarity of $e^{\mathrm{i}tH}$. The dynamics is then the familiar conjugation $A(t) = e^{\mathrm{i}tH} a e^{-\mathrm{i}tH}$, which we compute rather than postulate, and one more line, the role exchange of the observable and the probe, lets the bound be read from the observable's boundary as well as from the probe's, and the smaller reading taken.

The second half of the chapter builds the net on which spin systems live: a finite set of sites, each carrying a finite alphabet, the matrix algebra on the configurations, and the local algebra of a region, defined by what its matrices do to configurations rather than by a tensor product. The published bounds for spin systems are then corollaries, and a two-qubit example shows that the dynamics they bound is not trivial.

4.1 Hamiltonian families

Throughout the first half, $\mathcal{A}$ is a unital C*-algebra, nontrivial ($1 \neq 0$, so that a unitary has norm one), with a causal net $Z \mapsto \mathcal{A}(Z)$ over the subsets of a set of sites Λ (Definition 1.1). For $x \in \mathcal{A}$ write $\mathrm{L}_x : a \mapsto xa$ and $\mathrm{R}_x : a \mapsto ax$ for the left and right multiplications, bounded operators on $\mathcal{A}$ of norm at most $\|x\|$, and

$$\mathrm{ad}_x : a \longmapsto \mathrm{i}(xa - ax) = \mathrm{i}[x, a], \tag{4.1}$$

the *derivation* of x; ad is linear in x and $\|\mathrm{ad}_x\| \leq 2\|x\|$.

Definition 4.1 (Hamiltonian family)**.** A *Hamiltonian family* on the net is a finite family $(S_i, \Phi_i)_{i \in I}$ of finite supports $S_i \subseteq \Lambda$ and self-adjoint elements $\Phi_i \in \mathcal{A}(S_i)$. For $J \subseteq I$ put $H_J := \sum_{i \in J} \Phi_i$, and $H := H_I$. The generators of the family are $L_i := \mathrm{ad}_{\Phi_i}$, so that $L_J = \sum_{i \in J} L_i = \mathrm{ad}_{H_J}$ and $L = \mathrm{ad}_H$.

In the language of the sources the family is an *interaction*, a map from finite sets to self-adjoint local elements; we index it by a finite set so that one support may carry several terms, as in Chapter 1.

Proposition 4.2 (the derivation is a local generator)**.** Lean ✓ *Let $x \in \mathcal{A}(S)$ and $c \in \mathcal{A}$.*
(a) $\|\mathrm{ad}_x c\| \leq \|x\|\, N_S(c)$: locality in the argument holds with $\kappa = \|x\|$.
(b) If $c \in \mathcal{A}(Z)$ then $\mathrm{ad}_x c \in \mathcal{A}(Z \cup S)$: support closure.

Proof. (a) If $x = 0$ there is nothing to prove. Otherwise $b := x/\|x\|$ lies in $\mathcal{A}(S)$ with $\|b\| = 1$, and $\mathrm{ad}_x c = -\mathrm{i}\|x\|[c, b]$, so $\|\mathrm{ad}_x c\| = \|x\|\|[c, b]\| \leq \|x\|\, N_S(c)$ by the definition of the seminorm.
(b) By isotony both x and c lie in $\mathcal{A}(Z \cup S)$, which is closed under products, differences and scalars. □

The third hypothesis, contraction of every sub-family, is unitarity. To see it we compute the exponential of a derivation. The computation rests on a small fact about multiplications.

Proposition 4.3 (exponential of a multiplication)**.** Lean ✓ *For $x, y \in \mathcal{A}$: $\exp(\mathrm{L}_x) = \mathrm{L}_{\exp x}$, $\exp(\mathrm{R}_y) = \mathrm{R}_{\exp y}$, the operators L_x and R_y commute, and*

$$\exp(\mathrm{L}_x + \mathrm{R}_y)\, a = e^x\, a\, e^y \qquad (a \in \mathcal{A}). \tag{4.2}$$

Proof. $\mathrm{L}_x^k = \mathrm{L}_{x^k}$, and $x \mapsto \mathrm{L}_x$ is a bounded linear map, so it commutes with the convergent exponential series: $\exp(\mathrm{L}_x) = \sum_k \mathrm{L}_{x^k}/k! = \mathrm{L}_{\exp x}$. The same for R_y. Commutation is associativity, $(xa)y = x(ay)$. For commuting bounded operators P, Q on a Banach space, $\exp(P+Q) = \exp P \exp Q$ (the binomial rearrangement of the absolutely convergent series), and $\mathrm{L}_{e^x}\mathrm{R}_{e^y} a = e^x a e^y$. □

Proposition 4.4 (unitary conjugation). Lean ✓ *Let $H \in \mathcal{A}$ be self-adjoint and $t \in \mathbb{R}$. Then $U_t := \exp(\mathrm{i}tH)$ is unitary with $U_t^* = \exp(-\mathrm{i}tH) = U_{-t}$, and*

$$\exp(t\,\mathrm{ad}_H)\,a = U_t\,a\,U_t^*, \qquad \|\exp(t\,\mathrm{ad}_H)\,a\| = \|a\| \qquad (a \in \mathcal{A}). \tag{4.3}$$

Proof. The adjoint commutes with the exponential series, so $U_t^* = \exp((\mathrm{i}tH)^*) = \exp(-\mathrm{i}tH)$, and $\mathrm{i}tH$ commutes with $-\mathrm{i}tH$, so $U_t^* U_t = \exp(0) = 1 = U_t U_t^*$: U_t is unitary. On a, $t\,\mathrm{ad}_H a = \mathrm{i}tHa - a\,\mathrm{i}tH = (\mathrm{L}_{\mathrm{i}tH} + \mathrm{R}_{-\mathrm{i}tH})a$, and Proposition 4.3 gives $\exp(t\mathrm{ad}_H)a = e^{\mathrm{i}tH} a e^{-\mathrm{i}tH}$. Multiplication by a unitary on either side preserves the norm: $\|Ua\|^2 = \|a^*U^*Ua\| = \|a\|^2$ by the C*-identity, and $\|aU\| = \|(aU)^*\| = \|U^*a^*\| = \|a^*\| = \|a\|$ since U^* is unitary too. □

Theorem 4.5 (Hamiltonian families are local generators). Lean ✓ *A Hamiltonian family $(S_i, \Phi_i)_{i\in I}$ defines the family of local generators $(S_i, \mathrm{ad}_{\Phi_i}, \|\Phi_i\|)_{i\in I}$ of Definition 1.4, and its negation is contractive as well. The dynamics is*

$$A(t) = e^{tL}a = U_t\,a\,U_t^*, \qquad U_t = e^{\mathrm{i}tH}, \quad t \in \mathbb{R}. \tag{4.4}$$

Consequently the path-sum bound (2.4)*, the walk series* (3.5) *and the decay form* (3.9) *hold for every Hamiltonian family at every $t \in \mathbb{R}$ with $|t|$ in place of t (Theorem 3.16), with $\kappa_i = \|\Phi_i\|$.*

Proof. G1 and G2 are Proposition 4.2. For G3 fix J and $t \geq 0$: $L_J = \mathrm{ad}_{H_J}$ with H_J self-adjoint, so $\|e^{tL_J}a\| = \|a\|$ for all a by Proposition 4.4, that is $\|e^{tL_J}\| \leq 1$; at $-t$ the same gives G3 for the negated family. The formula is Proposition 4.4 with $J = I$. □

The other closed forms of Chapter 3, the matrix form on a finite lattice and the bounded-geometry and light-cone forms, hold for a Hamiltonian family as for every family of local generators for $t \geq 0$, and for every $t \in \mathbb{R}$ with $|t|$ in place of t by the paragraph after Theorem 3.16.

The sources define the dynamics through the Schrödinger equation of a propagator and prove unitarity from it; here the dynamics is e^{tL}, as in Chapter 2, and the conjugation form is a computation.

4.2 Role exchange

The path-sum bound treats the observable and the probe asymmetrically: the walks of Chapter 3 start on the boundary ∂Y of the probe's region. For an automorphism group we can exchange the two roles, and the bound improves to the smaller of the two readings. Theorem 3.1 of [6] carries this minimum, written $D(X, Y)$ in their eq. (3.24): the smaller of the sums of the decay function over $\partial Y \times X$ and over $\partial X \times Y$; their proof obtains the second from the same one-line identity, their eq. (3.39). Below, $B(t) := e^{tL}b$ is the evolved probe, and T, ∂, W_k are those of Chapter 3.

Theorem 4.6 (role exchange)**.** Lean ✓ *For a unitary U and $a, b \in \mathcal{A}$,*

$$[UaU^*, b] = U[a, U^*bU]U^*, \qquad \textit{hence} \qquad \|[UaU^*, b]\| = \|[a, U^*bU]\|. \tag{4.5}$$

For a Hamiltonian family, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ and $t \in \mathbb{R}$, $\|[A(t), b]\| = \|[B(-t), a]\|$, and

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \min\Big\{\sum_{k\ge1}\frac{(2|t|)^k}{k!}W_k(Y,X),\ \sum_{k\ge1}\frac{(2|t|)^k}{k!}W_k(X,Y)\Big\}\Big), \tag{4.6}$$

where $W_k(X, Y)$ sums the walks from ∂X to $Y \cap T$. The decay form of Theorem 3.9 holds likewise with $\min\{\Sigma_F(Y, X), \Sigma_F(X, Y)\}$.

Proof. $[UaU^*, b] = UaU^*b - bUaU^* = U(aU^*bU - U^*bUa)U^*$ since $U^*U = 1$; conjugation preserves the norm. With $U = U_t$, $U^*bU = U_{-t}bU_{-t}^* = B(-t)$ by (4.4). The first series is the time-reversed walk bound (Theorem 3.16); the second is the same bound at time $-t$ for the pair (b, a), whose regions are (Y, X), and $|-t| = |t|$. The smaller of two upper bounds is an upper bound; the minimum is of the two sums, not of their terms. □

4.3 Qudit lattices

Now let Λ be a finite set of sites, each carrying a finite nonempty alphabet Ω_u; a *qudit* of dimension n is the alphabet $\{0, \dots, n-1\}$ at every site. A *configuration* is a choice $s = (s(u))_{u\in\Lambda}$ of a letter at each site; there are finitely many. The *qudit algebra* M_Λ is the algebra of complex matrices indexed by configurations, with the operator norm of ℓ^2 over the configurations; it is a C*-algebra. For a region X write $s|_X$ for the restriction of s to X, and $s \sim_X s'$ when s and s' agree at every site outside X, that is, $s|_{X^c} = s'|_{X^c}$, and, for a choice ξ of letters on X, $(\xi, s|_{X^c})$ for the configuration equal to ξ on X and to s off X (Figure 6).

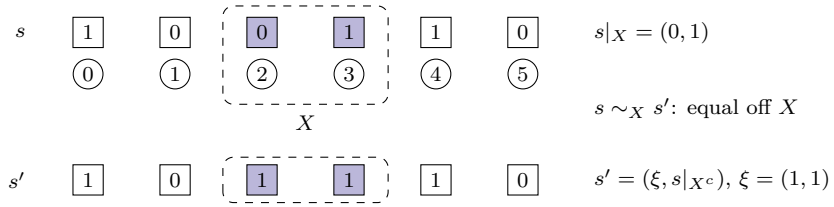


Figure 6: Two configurations of a chain of six sites with the alphabet $\{0, 1\}$, split by the region $X = \{2, 3\}$. They agree off X (the white boxes), so $s \sim_X s'$, and s' is $(\xi, s|_{X^c})$ for the letters $\xi = (1, 1)$ on X. A matrix local to X has a nonzero entry at (s, s') only for such pairs, and the entry depends on the shaded parts $s|_X$, $s'|_X$ alone (Definition 4.7).

Definition 4.7 (local algebra)**.** A matrix $M \in \mathrm{M}_\Lambda$ is *local to X* if

(1) $M_{ss'} = 0$ whenever $s \not\sim_X s'$, and

(2) $M_{ss'} = M_{rr'}$ whenever $s \sim_X s'$, $r \sim_X r'$, $s|_X = r|_X$ and $s'|_X = r'|_X$.

$\mathcal{A}(X)$ is the set of matrices local to X. For a matrix m on the letters of X, its *embedding* $\iota_X m$ is the matrix $(\iota_X m)_{ss'} := \mathbf{1}_{\{s\sim_X s'\}}\, m_{s|_X, s'|_X}$ (the tensor factor $m \otimes 1$ of the sources).

Condition (1) says that M does not change the letters outside X; condition (2) that what it does inside X does not depend on the letters outside. Together they say that M acts on the part inside X and as the identity outside, which is what $m \otimes 1$ means in the tensor-product picture; the definition avoids the tensor product and keeps every proof an entry computation. The embedded matrices are exactly the local ones.

Lemma 4.8 (sum over a class)**.** Lean ✓ *For a configuration s and a function f of configurations that vanishes off the class of s (that is, $f(s') = 0$ unless $s' \sim_X s$),*

$$\sum_{s'} f(s') = \sum_{\xi} f(\xi, s|_{X^c}), \tag{4.7}$$

the right side summing over the choices of letters on X.

Proof. $\xi \mapsto (\xi, s|_{X^c})$ is a bijection from the letters on X onto the class of s, with inverse $s' \mapsto s'|_X$; re-index the sum, dropping the vanishing terms. □

Proposition 4.9 (local algebras). Lean ✓
(a) $\mathcal{A}(X)$ is a star subalgebra of M_Λ containing 1, and it is closed.
(b) $\iota_X m \in \mathcal{A}(X)$, and every $M \in \mathcal{A}(X)$ is $\iota_X m$ for some matrix m on the letters of X.
(c) If $X \subseteq Y$ then $\mathcal{A}(X) \subseteq \mathcal{A}(Y)$; and $\mathcal{A}(\Lambda) = \mathrm{M}_\Lambda$.

Proof. (a) Both conditions are linear in M, hold for 1 (equal configurations agree off X, and for $s \sim_X s'$ the equality $s = s'$ is decided by $s|_X = s'|_X$), and are symmetric under $(s, s') \mapsto (s', s)$, which handles the adjoint. For the product let $M, N \in \mathcal{A}(X)$. If $s \not\sim_X s''$, every term $M_{ss'}N_{s's''}$ vanishes, since $M_{ss'} \neq 0$ forces $s' \sim_X s$ and $N_{s's''} \neq 0$ forces $s'' \sim_X s'$. If $s \sim_X s''$, Lemma 4.8 gives $(MN)_{ss''} = \sum_\xi M_{s,(\xi,s|_{X^c})}N_{(\xi,s|_{X^c}),s''}$, and each factor is unchanged when (s, s'') is replaced by (r, r'') with $r \sim_X r''$ and the same parts on X, by condition (2). Closedness: a subspace of a finite-dimensional space.

(b) The embedding satisfies (1) by its indicator and (2) because its entries depend on the parts on X only. Conversely, fix a configuration s_0, which exists as the alphabets are nonempty, and put $m_{\xi\eta} := M_{(\xi,s_0|_{X^c}),(\eta,s_0|_{X^c})}$. For $s \sim_X s'$ condition (2) with $r = (s|_X, s_0|_{X^c})$, $r' = (s'|_X, s_0|_{X^c})$ gives $M_{ss'} = m_{s|_X,s'|_X}$, and both sides vanish otherwise.

(c) If $s \not\sim_Y s'$ then $s \not\sim_X s'$. For (2), if $s \sim_X s'$ then $r \sim_X r'$, because off Y the pair (r, r') agrees and on $Y \setminus X$ it copies (s, s'), which agrees there; the restrictions to X agree, so condition (2) for X applies. If $s \not\sim_X s'$ then likewise $r \not\sim_X r'$ and both entries vanish. For $X = \Lambda$ both conditions are empty. □

Theorem 4.10 (the qudit net is causal). Lean ✓ *$X \mapsto \mathcal{A}(X)$ is a net over the subsets of Λ, and local matrices of disjoint regions commute: $M \in \mathcal{A}(X)$, $N \in \mathcal{A}(Y)$ and $X \cap Y = \emptyset$ give $MN = NM$.*

Proof. Fix s, s''. A term $M_{ss'}N_{s's''}$ is nonzero only if s' agrees with s off X and with s'' off Y; as $X \cap Y = \emptyset$ this determines s': on X it equals s'', elsewhere s. Call it $\bar{s}$; then $(MN)_{ss''} = M_{s\bar{s}}N_{\bar{s}s''}$. Likewise $(NM)_{ss''} = N_{s\hat{s}}M_{\hat{s}s''}$ with $\hat{s}$ equal to s'' on Y and to s elsewhere. If s and s'' differ somewhere off $X \cup Y$, then $\bar{s} \not\sim_Y s''$ and $\hat{s} \not\sim_X s''$, so both products vanish. Otherwise $M_{s\bar{s}} = M_{\hat{s}s''}$ by condition (2) for X ($s \sim_X \bar{s}$, $\hat{s} \sim_X s''$, $s|_X = \hat{s}|_X$, $\bar{s}|_X = s''|_X$) and $N_{\bar{s}s''} = N_{s\hat{s}}$ by (2) for Y, and the two products agree. □

Two further constructions serve the examples; both are for a uniform alphabet Ω. For a matrix m on Ω and a site u, the *single-site operator* m_u acts by m at the site u and trivially elsewhere, $(m_u)_{ss'} = \mathbf{1}_{\{s \sim_{\{u\}} s'\}} m_{s(u),s'(u)}$, the usual notation σ^x_u for a Pauli matrix at a site. It is local to $\{u\}$ (we read both conditions off the formula); $m_u m'_u = (mm')_u$, because the configurations agreeing with s off u are parametrized by their letter at u, so the sum over them is the matrix product at $(s(u), s''(u))$; operators at distinct sites commute by Theorem 4.10; and for $u \neq v$ and $s \sim_{\{u,v\}} s''$ the one nonzero term of the product gives $(m_u m'_v)_{ss''} = m_{s(u),s''(u)} m'_{s(v),s''(v)}$. A bijection $\varphi : \Lambda \to \Lambda'$ of site sets, both with the alphabet Ω, induces the star-isomorphism $M \mapsto M^\varphi$ of the qudit algebras, $(M^\varphi)_{rr'} := M_{r\circ\varphi, r'\circ\varphi}$ for configurations r, r' of Λ' (re-indexing along the bijection $r \mapsto r \circ \varphi$ of configurations, which commutes with products and adjoints); it carries $\mathcal{A}(X)$ onto $\mathcal{A}(\varphi(X))$, since $r \sim_{\varphi(X)} r'$ iff $r \circ \varphi \sim_X r' \circ \varphi$ and $r|_{\varphi(X)}$ determines $(r \circ \varphi)|_X$ and conversely, so that M is local to X iff M^φ is local to $\varphi(X)$, and φ^{-1} gives the inverse. This is the covariance a translation-invariant system on a finite torus needs. On two qubits, σ^x_1 is not local to $\{0\}$: the configurations 00 and 01 (the letter at the site 0 written first) differ at the site 1, outside $\{0\}$, yet the entry of σ^x_1 between them is 1, against condition (1). And σ^x_0, σ^z_0 do not commute: the net is not trivial.

4.4 The bounds for spin systems

Definition 4.11 (qudit interaction)**.** A *qudit interaction* is a finite family $(S_i, m_i)_{i\in I}$ of finite supports and Hermitian matrices m_i on the letters of S_i; its terms $\Phi_i := \iota_{S_i} m_i$ are self-adjoint and local to S_i, so the family is a Hamiltonian family on the qudit net.

Theorem 4.12 (the published bounds for qudits)**.** Lean ✓ *For a qudit interaction, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$, with the site matrix h of the strengths $\kappa_i = \|\Phi_i\|$:*

(i) for all $t \in \mathbb{R}$, the walk bound (4.6)*, and on the finite lattice*

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \sum_{y\in\partial Y}\sum_{x\in X\cap T}\big(e^{2|t|h} - 1\big)_{yx}\Big), \tag{4.8}$$

and the same with the roles exchanged, the sum over $\partial X \times (Y \cap T)$;

(ii) for a decay function (F, C_F) and an F-bound N_F, for all $t \in \mathbb{R}$,

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \frac{e^{2N_F C_F|t|} - 1}{C_F}\min\{\Sigma_F(Y, X), \Sigma_F(X, Y)\}\Big); \tag{4.9}$$

(iii) with bounded geometry, for $t \ge 0$ as stated there, the forms (3.10) *and* (3.11) *of Chapter 3.*

Proof. Theorem 4.5 puts the interaction, on the causal net of Theorem 4.10, under the hypotheses of Chapter 3 with a contractive negation; (i) and (ii) are the time-reversed forms with Theorem 4.6, the matrix form by Proposition 3.5 at $|t|$, (iii) the geometric forms. □

Equation (4.8) is the review's (3.36) [1], with its constant doubled and the -1 that the series proof gives (the review's $\mathcal{C}$ is half the supremum, and the identity has no entry between distinct sites), and with the sum restricted to $\partial Y \times (X\cap T)$; (4.9) is Theorem 3.1 of [6] for time-independent interactions, whose boundary ∂_Φ is ∂ when each support carries one nonzero term. Two further facts connect the decay functions of Chapter 3 to the sources' F-functions. In them, d is a distance on Λ as in Definition 3.10 ($d(u, u) = 0$ and the triangle inequality, not assumed symmetric), and T' ranges over the finite subsets of Λ.

Proposition 4.13 (the F-norm; weighted decay functions)**.** Lean ✓

(a) For a decay function $F > 0$ on the finite nonempty set Λ, $\|\Phi\|_F := \max_{x,y} h_{xy}/F(x, y)$ is an F-bound, and every F-bound N satisfies $N \ge \|\Phi\|_F$.

(b) If (F, C_F) is a decay function and d satisfies the triangle inequality, then for $\mu \ge 0$ the weighted decay function *$F_\mu(x, y) := e^{-\mu d(x,y)}F(x, y)$ is a decay function with the same constant C_F (the weighted F-functions of [6], eq. (3.11), for the weight μr).*

Proof. (a) A maximum dominates each value, and $F > 0$ clears the denominator; conversely $h_{xy}/F(x, y) \le N$ for every pair bounds the maximum.

(b) $e^{-\mu(d(x,z)+d(z,y))} \le e^{-\mu d(x,y)}$ for $\mu \ge 0$, so

$$\sum_{z\in T'} F_\mu(x, z)F_\mu(z, y) \le e^{-\mu d(x,y)} \sum_{z\in T'} F(x, z)F(z, y) \le C_F F_\mu(x, y). \qquad \square$$

With $F(x, y) = F_0(d(x, y))$ and one term per support, the number $\|\Phi\|_F$ of (a) is the F-norm $\sup_{x,y} F_0(d(x, y))^{-1}\sum_{Z\ni x,y}\|\Phi(Z)\|$ of [6], eq. (3.14); with several terms on a support, h adds their norms.

The polynomial decay function $F = F_0 \circ d$ with $F_0(r) = (1+r)^{-\alpha}$ on $\mathbb{Z}^\nu$ with the ℓ^1 distance has $C_F \le 2^\alpha\|F\|$ for $\alpha > \nu$, where $\|F\| = \sup_x \sum_y F(x, y)$; this is eq. (3.13) of [6], quoted here and not derived (the elementary estimate $(1+d(x, z))(1+d(z, y)) \ge \frac{1}{2}(1+d(x, y))(1+\min\{d(x, z), d(z, y)\})$ gives $2^{\alpha+1}\|F\|$). A finite piece of the lattice inherits the decay function with a constant at most that of the lattice. With (b) the weighted decay function F_μ gives the exponential light cone of (4.9) at the velocity $2N_{F_\mu}C_F/\mu$, with N_{F_μ} an F_μ-bound. On a chain of 20 sites at $\alpha = 2$ the least reproducing constant is 4.20 against $2^\alpha\|F\| = 8.43$; on a 6×6 grid at $\alpha = 3$ it is 4.46 against 16.3.

4.5 Two qubits

Example 4.14 (the exact commutator)**.** script ✓ Two qubits, with the Pauli matrices $\sigma^x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma^y = \left(\begin{smallmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{smallmatrix}\right)$, $\sigma^z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ at the sites, σ^x_u and so on. Let $H = \sigma^x_0\sigma^x_1$ (the single term with support $\{0,1\}$ and norm 1), $a = \sigma^z_0$, $b = \sigma^z_1$. Since $H^2 = 1$ and H anticommutes with a,

$$A(t) = e^{\mathrm{i}tH} a e^{-\mathrm{i}tH} = a\, e^{-2\mathrm{i}tH} = a(\cos 2t - \mathrm{i}\sin 2t\, H), \qquad [A(t), b] = -2\mathrm{i}\sin 2t\ \sigma^y_0\sigma^y_1,$$

so $\|[A(t), b]\| = 2|\sin 2t|$. The bound (4.8) has $h = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $\partial Y = \{1\}$, $X \cap T = \{0\}$ and $(e^{2th} - 1)_{10} = (e^{4t} - 1)/2$, hence $\|[A(t), b]\| \le e^{4t} - 1$: the ratio of the bound to the exact norm is 1.24 at $t = 0.1$, 2.05 at $t = 0.3$ and 11.1 at $t = \pi/4$, where the commutator is largest. The bound is a bound on growth, not on the oscillation that follows. The system is that of Example 1.9 with σ^x and σ^z exchanged; as there, the path-sum bound of Chapter 2 gives $4t$ for it, while the walk bound gives $e^{4t} - 1$, the price of counting walks in place of paths.

A second route to the same conclusion, that the dynamics bounded in this chapter is not trivial, avoids the exponential: at $t = 0$ the commutator vanishes by causality, and its derivative there is $[\mathrm{i}[H, a], b] = \mathrm{i}\, c_0 c_1$, the product of the single-site operators of the matrix $c = \sigma^x\sigma^z - \sigma^z\sigma^x = -2\mathrm{i}\sigma^y$ at the sites 0 and 1. By the product formula for single-site operators, its entry at the row configuration 00 and the column configuration 11 is $\mathrm{i}\, c(0,1)\, c(0,1) = 4\mathrm{i} \neq 0$, where $c(0,1)$ is the entry of the 2×2 matrix c in the row 0 and the column 1. A function vanishing at 0 with a nonzero derivative there is nonzero at some $t > 0$; hence there is $t > 0$ with $[A(t), b] \neq 0$.

Notes and references

The review's bound for spin systems. For edge Hamiltonians H_e on a graph, the review [1] puts $h_{uv} = \sum_{e \ni u,v} \|H_e\|$ (its eq. (3.35)) and writes $\mathcal{C}_{AB}(t)$ for half the supremum of the commutator norm of an evolved unit operator on the region A with a unit operator on the region B (its eq. (3.24)); its bound (3.36) reads

$$\mathcal{C}_{AB}(t) \le \sum_{u \in A,\, v \in B} \left(e^{2|t|h}\right)_{uv}.$$

Theorem 4.12(i) gives it. Take the edges as the supports of a qudit interaction with the terms H_e, so that $\kappa_e = \|H_e\|$ and h is the review's matrix, and put $X = A$, $Y = B$. Halving (4.8) at unit a, b gives $\mathcal{C}_{AB}(t) \le \mathbf{1}_{\{A \cap B \neq \emptyset\}} + \sum_{y \in \partial B} \sum_{x \in A \cap T} (e^{2|t|h} - 1)_{yx}$. The entries of $e^{2|t|h} - 1 = \sum_{k \ge 1} (2|t|h)^k / k!$ are nonnegative and h is symmetric, so the double sum is at most $\sum_{u \in A, v \in B} (e^{2|t|h} - 1)_{uv}$, and the indicator is at most the sum of the entries of the identity over $A \times B$; together these give (3.36).

Theorem 3.1 of [6]. For an interaction Φ on a metric space of sites, an F-function F with constant C_F and the F-norm $\|\Phi\|_F$, the theorem states, for a time-independent interaction, $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$ and $X \cap Y = \emptyset$,

$$\|[\tau_t(A), B]\| \le \frac{2\|A\|\|B\|}{C_F} \left(e^{2\|\Phi\|_F C_F |t|} - 1\right) D(X, Y),$$

with $D(X, Y)$ the smaller of $\sum_{x \in X} \sum_{y \in \partial_\Phi Y} F(d(x, y))$ and $\sum_{x \in \partial_\Phi X} \sum_{y \in Y} F(d(x, y))$ (their eq. (3.24)). Theorem 4.12(ii) gives it on a finite set of sites: take one term $\Phi(Z)$ per support, the decay function $(x, y) \mapsto F(d(x, y))$, for which their C_F is a reproducing constant since finite sums are at most the full ones, and the F-bound $\|\Phi\|_F$ (Proposition 4.13(a) and the paragraph after it). The boundary ∂ is their ∂_Φ, and $\Sigma_F(Y, X)$, $\Sigma_F(X, Y)$ are at most the two sums of $D(X, Y)$, since they run over the touched sites only and their d is symmetric.

Checks. The algebra of the local matrices and the embedding, the conjugation form of the dynamics, the two-qubit example (its exact norm, the entries of the double commutator and the ratios above), the role exchange on a random three-site Hamiltonian, the F-norm and the weighted decay functions on chains, and the reproducing constants of the polynomial decay function on the chain of 20 sites and the 6×6 grid are computed by `checks/m2_check.py`.

5 Local approximation and open systems

The bounds of the previous chapters control the commutator of an evolved observable $A(t)$ with every observable of a distant region. Physically one wants a stronger-looking statement: that $A(t)$ *is*, up to a small error, an observable of a bounded region, one that grows at the velocity of the bound. The two statements are equivalent up to constants, and the device that converts one into the other is a conditional expectation P_X onto the local algebra of a region: a linear map that projects every observable onto its part living in X. This chapter builds P_X on the qudit lattice as a finite average over unitaries of the complement, identifies it, in the tensor-product picture, with the normalized partial trace, and proves that the distance from M to $\mathcal{A}(X)$ is controlled by the commutators of M with the complement, with constants 1 and 2. The evolved observable is then within a light-cone tail of a local one, and the dynamics is an *approximate cellular automaton*: it maps the observables of O into those of a neighbourhood of O, up to an error.

The second theme is dissipation. A quantum system coupled to an environment evolves, in the Heisenberg picture, under a Lindblad generator: a Hamiltonian part and jump terms. Such a generator built from local pieces satisfies the hypotheses of the path-sum bound, with the contraction of its sub-families, a consequence of complete positivity, carried as the one hypothesis not reproved here: locality in the argument comes from a one-line commutator identity, and the conditional expectation gives a second route to it with the norm of the term as strength. The bound and its closed forms for $t \geq 0$ therefore hold for open systems with no new analysis.

5.1 Nets with local approximation

Throughout, $\mathcal{A}$ is a unital C*-algebra, nontrivial where a unitary has to have norm one, with a net $Z \mapsto \mathcal{A}(Z)$ over the subsets of a set of sites Λ, causal where stated (Definition 1.1), and $N_S(c)$ is the commutator seminorm of Chapter 1.

Definition 5.1 (local approximation)**.** The net has *local approximation* if every region X carries a map $P_X : \mathcal{A} \to \mathcal{A}$ with

$$P_X c \in \mathcal{A}(X) \qquad \text{and} \qquad \|c - P_X c\| \leq N_{X^c}(c) \qquad (c \in \mathcal{A}). \tag{5.1}$$

A map $\mathcal{T} : \mathcal{A} \to \mathcal{A}$ is *ε-local with the neighbourhood map $\mathcal{N}$* (a map from regions to regions) if for every region O and every $c \in \mathcal{A}(O)$ there is $c' \in \mathcal{A}(\mathcal{N}(O))$ with $\|\mathcal{T}c - c'\| \leq \varepsilon \|c\|$.

The inequality in (5.1) says that an element commuting with everything outside X up to ε is within ε of an element of $\mathcal{A}(X)$: a statement about commutators becomes a statement about distance to a subalgebra. At $X = \Lambda$ it forces $P_\Lambda c = c$ whenever $\mathcal{A}(\emptyset)$ consists of scalars, so a net with local approximation has $\mathcal{A}(\Lambda) = \mathcal{A}$, as the qudit net does. The converse holds on every causal net, with the factor 2 of the commutator.

Proposition 5.2 (the converse estimate; the exact case)**.** Lean ✓

(a) On a causal net, if $c' \in \mathcal{A}(X)$ and Y is disjoint from X, then $N_Y(c) \le 2\|c - c'\|$ for every c. On a causal net with local approximation, $N_{X^c}(c) \le 2\|c - P_X c\|$.

(b) $\mathcal{T}$ is 0-local with $\mathcal{N}$ if and only if $\mathcal{T}(\mathcal{A}(O)) \subseteq \mathcal{A}(\mathcal{N}(O))$ for every O; this is the locality condition of a quantum cellular automaton with the neighbourhood $\mathcal{N}(O)$.

Proof. (a) For $b \in \mathcal{A}(Y)$ with $\|b\| \le 1$, c' and b commute by causality, so $[c, b] = [c - c', b]$ and $\|[c, b]\| \le 2\|c - c'\|$; take the supremum over b.

(b) At $\varepsilon = 0$ the condition reads $\mathcal{T}c = c'$ for some $c' \in \mathcal{A}(\mathcal{N}(O))$. □

5.2 The conditional expectation on the qudit lattice

Let Λ be finite with finite nonempty alphabets Ω_u, and M_Λ the qudit algebra of Chapter 4. For a region X write Ω_{X^c} for the letters on the complement and, for $\eta \in \Omega_{X^c}$ and a configuration s, $(s|_X, \eta)$ for the configuration with the letters of s on X and η off X.

Definition 5.3 (conditional expectation)**.** The *conditional expectation* onto X is the linear map P_X of M_Λ with

$$(P_X M)_{ss'} := \mathbf{1}_{\{s \sim_X s'\}} \, \frac{1}{|\Omega_{X^c}|} \sum_{\eta \in \Omega_{X^c}} M_{(s|_X, \eta),\,(s'|_X, \eta)}. \tag{5.2}$$

In the tensor-product picture this is the normalized partial trace over the complement, re-embedded as $|\Omega_{X^c}|^{-1} \operatorname{tr}_{X^c} M \otimes 1$: the entry at (s, s') averages the entries of M between the pairs that agree with s, s' on X and carry the same letters off X. The formula makes sense for any finite alphabets.

Proposition 5.4 (properties of the conditional expectation)**.** Lean ✓

(a) $P_X M$ is local to X, and $P_X M = M$ when M is local to X; so $P_X \circ P_X = P_X$.

(b) P_X is linear, $P_X 1 = 1$ and $(P_X M)^* = P_X(M^*)$.

(c) $P_\Lambda = \mathrm{id}$ and $P_\emptyset M = \frac{\operatorname{tr} M}{|\Omega_\Lambda|}\, 1$.

Proof. (a) The entry (5.2) vanishes unless $s \sim_X s'$ and depends on s, s' through $s|_X$, $s'|_X$: the two conditions of Definition 4.7. If M is local to X and $s \sim_X s'$, every term of the sum equals $M_{ss'}$, because $(s|_X, \eta) \sim_X (s'|_X, \eta)$ with the same restrictions to X as s, s'; the average of $|\Omega_{X^c}|$ copies of $M_{ss'}$ is $M_{ss'}$. If $s \not\sim_X s'$ both sides vanish.

(b) Linearity is termwise; 1 is local to every region; the conjugate of the average is the average of the conjugates, and $s \sim_X s'$ is symmetric.

(c) For $X = \Lambda$ there is one letter η off X, the empty one, and $(s|_\Lambda, \eta) = s$. For $X = \emptyset$ the condition $s \sim_\emptyset s'$ is $s = s'$ and η runs over all configurations. □

Two properties remain: that P_X is a contraction, and the estimate (5.1). Both come from a second expression of P_X, as an average of unitary conjugations, which needs a group structure on the alphabets. From here on each alphabet is cyclic, $\Omega_u = \mathbb{Z}_{q_u}$ with $q_u \ge 1$ (every finite alphabet is one up to a renaming of its letters), and we add letters modulo q_u at each site. Write $e_n(k) := e^{2\pi \mathrm{i} k/n}$ for $k \in \mathbb{Z}_n$; it is a character, $e_n(k + k') = e_n(k) e_n(k')$, with $e_n(k)^n = 1$ and $\overline{e_n(k)} = e_n(-k)$.

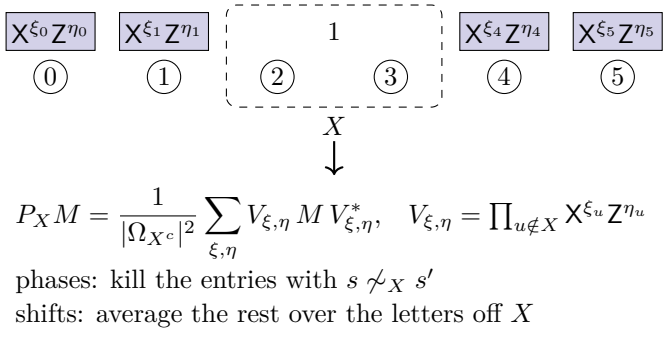


Figure 7: The twirl onto $X = \{2, 3\}$ on a chain of six sites. Each site off X carries a shift power and a clock power, with the letters ξ_u, η_u of that site; the sites of X carry the identity. Averaging $V_{\xi,\eta} M V_{\xi,\eta}^*$ over all letters off X gives the conditional expectation (5.2): the clock phases cancel every entry between configurations that differ off X, and the shifts average the remaining entries over the letters off X (Theorem 5.8).

Proposition 5.5 (characters, clock and shift)**.** Lean ✓ *For $n \geq 1$ and $k \in \mathbb{Z}_n$,*

$$\sum_{l\in\mathbb{Z}_n} e_n(lk) = n\, \mathbf{1}_{\{k=0\}}. \tag{5.3}$$

The clock Z *and the* shift X *are the $n \times n$ matrices $\mathsf{Z}_{ll'} = e_n(l)\mathbf{1}_{\{l=l'\}}$, $\mathsf{X}_{ll'} = \mathbf{1}_{\{l=l'+1\}}$; they satisfy $\mathsf{X}\mathsf{X}^* = 1$ and $\mathsf{Z}\mathsf{X} = e_n(1)\,\mathsf{X}\mathsf{Z}$.*

Proof. If $k = 0$ every term of (5.3) is 1. Otherwise $e_n(k) \neq 1$ is an n-th root of unity, so the geometric sum $\sum_{l<n} e_n(k)^l = (e_n(k)^n - 1)/(e_n(k) - 1)$ vanishes. The shift is a permutation matrix, so $\mathsf{X}\mathsf{X}^* = 1$, and $(\mathsf{Z}\mathsf{X})_{ll'} = e_n(l)\mathbf{1}_{\{l=l'+1\}} = e_n(1)e_n(l')\mathbf{1}_{\{l=l'+1\}} = e_n(1)(\mathsf{X}\mathsf{Z})_{ll'}$. □

Both matrices are unitary: the clock is diagonal with unimodular entries, and the shift, a permutation matrix, satisfies $\mathsf{X}^*\mathsf{X} = 1$ as well.

On a region Y the products of clock and shift powers over the sites of Y form the *Weyl unitaries*; we define them directly by their entries, which is what every proof uses (Figure 7). The average of the conjugations by the Weyl unitaries of the complement of X is a *twirl*, the standard name in quantum information for an average of conjugations over a group of unitaries; it will be P_X. For letters $\xi \in \Omega_Y$ and a configuration s, $s + \xi$ is the configuration with $s(u) + \xi(u)$ on Y and $s(u)$ off Y, and $s - \xi$ likewise; $s - s'$ is the sitewise difference of two configurations, of which the phase ψ_η sees the letters on Y only.

Definition 5.6 (Weyl unitaries, twirl)**.** For a region Y and letters $\xi, \eta \in \Omega_Y$, the *Weyl unitary* $V_{\xi,\eta}$ is the matrix

$$(V_{\xi,\eta})_{ss'} := \mathbf{1}_{\{s=s'+\xi\}}\ \psi_\eta(s'), \qquad \psi_\eta(s) := \prod_{u\in Y} e_{q_u}(\eta(u)\, s(u)), \tag{5.4}$$

the shift by ξ applied after the clock phases ψ_η. The *twirl* onto X is the average over the Weyl unitaries of the complement $Y = X^c$,

$$\frac{1}{|\Omega_Y|^2} \sum_{\xi\in\Omega_Y} \sum_{\eta\in\Omega_Y} V_{\xi,\eta}\, M\, V_{\xi,\eta}^*. \tag{5.5}$$

Proposition 5.7 (Weyl unitaries)**.** Lean ✓ *$V_{\xi,\eta}$ is local to Y and unitary, and for every M and configurations s, s'*

$$(V_{\xi,\eta} M V_{\xi,\eta}^*)_{ss'} = \psi_\eta(s - s')\ M_{s-\xi,\, s'-\xi}. \tag{5.6}$$

Proof. The phase $\psi_\eta(s)$ depends on $s|_Y$ only and is multiplicative in s, with $\overline{\psi_\eta(s)} = \psi_\eta(-s)$ and $|\psi_\eta| = 1$. The entry (5.4) vanishes unless $s \sim_Y s'$, since $s' + \xi \sim_Y s'$, and depends on the restrictions to Y alone; so $V_{\xi,\eta}$ is local to Y. In V^*V and VV^* the sums over the middle configuration have one term, at $s+\xi$ and at $s-\xi$ respectively, where the phase meets its conjugate; both products are the identity. In $(VMV^*)_{ss'} = \sum_{r,r'} V_{sr} M_{rr'} \overline{V_{s'r'}}$ the only term is $r = s - \xi$, $r' = s' - \xi$, with the phase $\psi_\eta(s-\xi)\overline{\psi_\eta(s'-\xi)} = \psi_\eta(s-s')$, the ξ cancelling. □

Theorem 5.8 (the twirl is the conditional expectation)**.** Lean ✓ *For cyclic alphabets the twirl* (5.5) *onto X equals $P_X M$ for every M.*

Proof. By (5.6) the entry of the twirl at (s, s') is

$$\frac{1}{|\Omega_Y|^2} \sum_{\xi} M_{s-\xi,s'-\xi} \sum_{\eta} \psi_\eta(s-s'), \qquad \text{and} \qquad \sum_{\eta} \psi_\eta(s-s') = \prod_{u \in Y} \sum_{l \in \mathbb{Z}_{q_u}} e_{q_u}\big(l\,(s(u) - s'(u))\big)$$

equals $|\Omega_Y|\,\mathbf{1}_{\{s \sim_X s'\}}$ by (5.3) at each site. For $s \sim_X s'$ the letters of s and s' off X agree, so $s - \xi = (s|_X, \zeta)$ and $s' - \xi = (s'|_X, \zeta)$ with $\zeta := s|_Y - \xi$, and as ξ runs over Ω_Y so does ζ. The remaining sum is that of (5.2), with the normalization $|\Omega_Y|/|\Omega_Y|^2 = 1/|\Omega_Y|$. □

Proposition 5.9 (contraction and bimodule property)**.** Lean ✓ *For cyclic alphabets, $\|P_X M\| \le \|M\|$, and $P_X(aMb) = a\,(P_X M)\,b$ for $a, b \in \mathcal{A}(X)$.*

Proof. Each term $V_{\xi,\eta} M V_{\xi,\eta}^*$ of the twirl has the norm of M, and the norm of an average is at most the average of the norms. For $a, b \in \mathcal{A}(X)$ and V local to X^c, $V(aMb)V^* = a(VMV^*)b$ by causality (Theorem 4.10), termwise in the average. □

Theorem 5.10 (local approximation)**.** Lean ✓ *For cyclic alphabets, every $M \in \mathrm{M}_\Lambda$ and every region X,*

$$\|M - P_X M\| \le N_{X^c}(M) \le 2\,\|M - P_X M\|. \tag{5.7}$$

The qudit net with cyclic alphabets has local approximation with P_X.

Proof. Since $VV^* = 1$, $M - VMV^* = [M, V]\,V^*$ for each Weyl unitary V of the complement, so $M - P_X M$ is the average of the $[M, V]\,V^*$. Each has norm at most $\|[M, V]\| \le N_{X^c}(M)$, because $V \in \mathcal{A}(X^c)$ has norm one; so does the average. The second inequality is Proposition 5.2(a) with $c = M$ and $c' = P_X M \in \mathcal{A}(X)$. □

The constant 2 is attained: for a single qubit outside X and $M = \sigma^x$ at that site, $P_X M = 0$, $\|M - P_X M\| = 1$, and $N_{X^c}(M) = 2$ (at least $\|[\sigma^x, \sigma^z]\| = 2$, at most $2\|M\|$). The constant 1 is approached along the following family: for $X = \emptyset$, one site with an alphabet of $n \ge 2$ letters and M a rank-one projection $|e\rangle\langle e|$, $\|M - P_\emptyset M\| = 1 - 1/n$ while $N_\Lambda(M) = 1$ (the commutator of a projection with an element of norm at most one has two off-diagonal blocks of norm at most one, and $|e\rangle\langle f| + |f\rangle\langle e|$ with $f \perp e$ gives one).

5.3 Truncated dynamics and the approximate automaton

Let $(S_i, L_i, \kappa_i)_{i \in I}$ be a family of local generators on a causal net with local approximation, $A(t) = e^{tL} a$ for $a \in \mathcal{A}(X)$, and let $X' \supseteq X$ be a larger region. The walk bound of Chapter 3 controls the commutator of $A(t)$ with every observable of $Y := X'^c$; local approximation turns this into a distance.

Theorem 5.11 (truncated dynamics). Lean ✓ *For $a \in \mathcal{A}(X)$, $X \subseteq X'$, $Y = X'^c$ and $t \geq 0$, with the walk terms $\omega_k(t)$ of Chapter 3 for the pair (Y, X), eq.* (3.6)*,*

$$\|A(t) - P_{X'}A(t)\| \leq N_Y(A(t)) \leq 2\|a\| \sum_{k\geq 0} \omega_k(t). \tag{5.8}$$

Under bounded geometry (g, ℓ, p, ρ) with distance d and $v = 2g\ell p$, if $(r-1)\rho < d(y,x)$ for all $y \in \partial Y$ and $x \in X \cap T$ (an integer $r \geq 1$), the right side is at most $2\|a\|\,|\partial Y|\,(vt)^r e^{vt}/r!$ and, for every real μ, at most $2\|a\|\,|\partial Y|\,e^{vt}e^{-\mu r}(e^{e^\mu vt} - 1)$.

Proof. The first inequality is local approximation. For the second, $N_Y(A(t))$ is the supremum of $\|[A(t), b]\|$ over $b \in \mathcal{A}(Y)$ with $\|b\| \leq 1$, and the walk bound (Theorem 3.4) gives $2\|a\|\|b\|\,(\mathbf{1}_{\{X\cap Y\neq\emptyset\}} + \sum_k \omega_k(t))$ with $X \cap Y = \emptyset$. The geometric forms are those of Chapter 3 for the walk series. □

So $P_{X'}A(t)$ is an observable of X' within a light-cone tail of $A(t)$. Read as a property of the map e^{tL}, this is approximate locality with a neighbourhood map. The natural map is the r-neighbourhood $O^{+r} := \{w : d(w,x) < r\rho \text{ for some } x \in O\}$, measured from the site into the region (the distance need not be symmetric).

Corollary 5.12 (approximate automaton). Lean ✓

(a) For Λ finite and a neighbourhood map $\mathcal{N}$ with $O \subseteq \mathcal{N}(O)$ for all O, e^{tL} is $\varepsilon(t)$-local with $\mathcal{N}$, where $\varepsilon(t)$ is the largest, over the finitely many regions O, of the errors $2\sum_k \omega_k(t)$ of the pairs $(\mathcal{N}(O)^c, O)$.

(b) Under bounded geometry with $\rho > 0$ and an integer $r \geq 1$, e^{tL} is $\varepsilon_r(t)$-local with $O \mapsto O^{+r}$, where $\varepsilon_r(t)$ is either of

$$\varepsilon_r(t) = 2|T|\,\frac{(vt)^r}{r!}e^{vt}, \tag{5.9}$$

$$\varepsilon_r(t) = 2|T|\,e^{vt}e^{-\mu r}(e^{e^\mu vt} - 1) \qquad (\mu \in \mathbb{R}). \tag{5.10}$$

Proof. (a) For $c \in \mathcal{A}(O)$ take $P_{\mathcal{N}(O)}e^{tL}c$ as the nearby element of Definition 5.1 and apply (5.8) with $X = O$, $X' = \mathcal{N}(O)$; the maximum over O makes the error uniform.

(b) $O \subseteq O^{+r}$ since $d(u,u) = 0 < r\rho$. A site $w \notin O^{+r}$ has $d(w,u) \geq r\rho > (r-1)\rho$ for every $u \in O$, so the geometric form applies, and $|\partial Y| \leq |T|$ because the boundary consists of touched sites. □

At fixed t the first error of (b) vanishes as $r \to \infty$, since $(vt)^r/r! \to 0$, and the second, for $\mu > 0$, is at most $2|T|\,e^{(1+e^\mu)vt - \mu r}$, so exponentially small in r once $r > (1+e^\mu)vt/\mu$: up to time t the dynamics is a cellular automaton with the neighbourhood O^{+r}, r of order vt, up to an exponentially small error.

Example 5.13 (truncation on a chain). script ✓ On a chain of five qubits with $H = \sum_{u=0}^{3}(\sigma_u^z\sigma_{u+1}^z + \frac{1}{2}\sigma_u^x\sigma_{u+1}^x)$ and $a = \sigma_0^x$, the error $\|A(t) - P_{X_r}A(t)\|$ for $X_r = \{0, \ldots, r\}$ against the walk bound (5.8): at $t = 0.3$ the error is 0.56, 0.086, 0.017 for $r = 0, 1, 2$, against the bounds 9.6, 4.6, 1.4; at $t = 0.1$ and $r = 2$ the error is $6.6 \cdot 10^{-4}$ against $1.6 \cdot 10^{-2}$. The error decays with r as the bound predicts; the bound overshoots by factors between 17 and 82 at these times, and the first two bounds quoted exceed the trivial bound $2\|a\|$. For $X_r = \Lambda$ the error is zero, as $A(t)$ is then local to X_r.

On twelve qubits with the same Hamiltonian and observable the light cone is visible (Figure 8). At $t = 0.25$, 0.5, 1, 2 the error falls below 0.1 from $r = 1$, 2, 4, 7 on: a front moving at about 3.5 sites per unit time. Beyond the front the error decays by a constant factor per site, as $e^{-3.07r}$, $e^{-2.38r}$, $e^{-1.66r}$, $e^{-0.86r}$ at the four times (least squares over $r = 3, \ldots, 10$), the rate falling as the front approaches.

The walk bound (5.8) decays in r in the same way, at the smaller rates $e^{-2.34r}$, $e^{-1.67r}$, $e^{-1.05r}$, $e^{-0.57r}$; it is below the trivial bound $2\|a\|$ from $r = 2$ on at $t = 0.25$ and from $r = 5$ on at $t = 0.5$, and trivial for every $r \le 10$ at $t = 1$ and $t = 2$, where its velocity has outrun the chain. It exceeds the error by factors from 12 to above 10^{10}. The bounded-geometry bound with $v = 16$ is trivial throughout. Corollary 5.12 and the paragraph after it predict the exponential decay in the radius and a front moving linearly in t; the velocity they give exceeds the observed one.

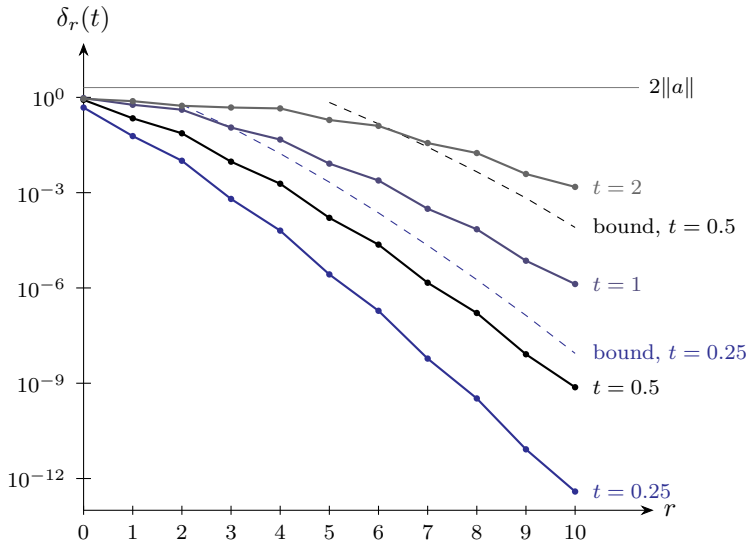


Figure 8: The truncation error $\delta_r(t) = \|A(t) - P_{X_r}A(t)\|$ on a chain of twelve qubits, $H = \sum_u(\sigma^z_u\sigma^z_{u+1} + \frac{1}{2}\sigma^x_u\sigma^x_{u+1})$, $a = \sigma^x_0$, $X_r = \{0, \dots, r\}$, against r on a logarithmic scale, one curve per time t. Dashed: the walk bound (5.8) where it is below the trivial bound $2\|a\|$ (at $t = 1$ and 2 it is trivial for every r shown). Beyond a front that moves with t the error decays exponentially in r; the bound shares the decay, with a smaller rate and a far larger prefactor.

5.4 Open systems

A quantum system in contact with an environment evolves, after the environment has been traced out and under the Markov approximation, by a semigroup e^{tL} of unital completely positive maps, whose generator has the Lindblad form. In the Heisenberg picture, for a Hamiltonian H and finitely many *jump operators* Γ_j,

$$La = \mathrm{i}[H, a] + \sum_j \Big(\Gamma_j^* a \Gamma_j - \tfrac{1}{2}\big(\Gamma_j^*\Gamma_j a + a\Gamma_j^*\Gamma_j\big)\Big). \tag{5.11}$$

The jump terms describe the environment's action: a measurement-like kick $a \mapsto \Gamma^* a \Gamma$ balanced by the anticommutator that keeps $L1 = 0$. On a lattice the Hamiltonian and the jumps are local, and the generator is a sum of local terms.

Definition 5.14 (Lindblad terms and families)**.** A *Lindblad term* on a finite region S is a self-adjoint $H \in \mathcal{A}(S)$ with finitely many jump operators $\Gamma_j \in \mathcal{A}(S)$; its generator L_S is (5.11), and the *dissipator* of one jump is $\Delta_\Gamma a := \Gamma^* a \Gamma - \frac{1}{2}(\Gamma^*\Gamma a + a\Gamma^*\Gamma)$, so that $L_S = \mathrm{ad}_H + \sum_j \Delta_{\Gamma_j}$ (with no jumps the term is a Hamiltonian one). A *Lindblad family* is a finite family $(S_i, L_i)_{i\in I}$ of Lindblad terms, with Hamiltonian parts H_i and jumps Γ_{ij}, finitely many j for each i, together with the hypothesis that every sub-family is *contractive*: $\|e^{tL_J}\| \le 1$ for all $J \subseteq I$ and $t \ge 0$.

The contraction hypothesis is physics: e^{tL_J} is unital and completely positive, and a unital positive map between C*-algebras has norm one. We reprove neither that norm theorem nor the product formula that passes from the terms to their sum; we carry the hypothesis explicitly and prove one case, a single dephasing jump, below (Proposition 5.18).

Proposition 5.15 (Lindblad terms). Lean ✓ *For a Lindblad term on S and a causal net:*

(a) L_S *is bounded,* $\|L_S\| \le 2\|H\| + 2\sum_j \|\Gamma_j\|^2$*;*

(b) $L_S c = 0$ *for* $c \in \mathcal{A}(S^c)$*;*

(c) $L_S c \in \mathcal{A}(Z \cup S)$ *for* $c \in \mathcal{A}(Z)$ *(support closure, G2);*

(d) $(L_S a)^* = L_S(a^*)$.

Proof. (a) $\|\mathrm{ad}_H a\| \le 2\|H\|\|a\|$ by (4.1); the three maps $a \mapsto \Gamma^* a \Gamma$, $\frac{1}{2}\Gamma^*\Gamma a$, $\frac{1}{2} a \Gamma^* \Gamma$ have norms at most $\|\Gamma\|^2$, $\frac{1}{2}\|\Gamma\|^2$, $\frac{1}{2}\|\Gamma\|^2$.

(b) c commutes with H, Γ_j and Γ_j^* by causality ($\mathcal{A}(S)$ is closed under adjoints), so $[H, c] = 0$ and $\Gamma_j^* c \Gamma_j = \Gamma_j^* \Gamma_j c = \frac{1}{2}(\Gamma_j^*\Gamma_j c + c\Gamma_j^*\Gamma_j)$.

(c) Every factor lies in $\mathcal{A}(Z \cup S)$ by isotony, which is closed under products and sums.

(d) $(\mathrm{i}[H, a])^* = \mathrm{i}[H, a^*]$ for self-adjoint H, $(\Gamma^* a \Gamma)^* = \Gamma^* a^* \Gamma$, and the two halves of the anticommutator exchange. □

Theorem 5.16 (locality in the argument for Lindblad terms). Lean ✓ *For a Lindblad term on S and every c:*

(a) on any net, $\|\Delta_\Gamma c\| \le \|\Gamma\|^2 N_S(c)$ *for* $\Gamma \in \mathcal{A}(S)$*, hence*

$$\|L_S c\| \le (\|H\| + \textstyle\sum_j \|\Gamma_j\|^2) N_S(c); \tag{5.12}$$

(b) on a causal net with local approximation, $L_S c = L_S(c - P_{S^c} c)$ *and*

$$\|L_S c\| \le \|L_S\|\, N_S(c) \le \big(2\|H\| + 2\textstyle\sum_j \|\Gamma_j\|^2\big) N_S(c). \tag{5.13}$$

Proof. (a) The dissipator is a sum of two commutators, $\Delta_\Gamma c = \frac{1}{2}\Gamma^*[c, \Gamma] + \frac{1}{2}[\Gamma^*, c]\,\Gamma$ (expand both sides), and $\|[c, x]\| \le \|x\|\, N_S(c)$ for $x \in \mathcal{A}(S)$ (Proposition 4.2), applied to $x = \Gamma$ and $x = \Gamma^*$; the Hamiltonian part is Proposition 4.2 again.

(b) $P_{S^c} c \in \mathcal{A}(S^c)$, so $L_S(P_{S^c} c) = 0$ by Proposition 5.15(b) and $L_S c = L_S(c - P_{S^c} c)$ by linearity. Then $\|L_S c\| \le \|L_S\|\|c - P_{S^c} c\| \le \|L_S\| N_S(c)$ by (5.1) for the region S^c, whose complement is S. □

Route (a) is the sharper one: it needs neither causality nor the conditional expectation, and with no jumps it returns the Hamiltonian strength $\|H\|$ of Chapter 4. Route (b) is the mechanism in one line: a term acts only on what an observable does inside its support, the conditional expectation isolates that part, and the strength is the norm of the term, the form in which the literature states the open-system bound.

Theorem 5.17 (open systems as local generators). Lean ✓ *On a causal net, a Lindblad family $(S_i, L_i)_{i \in I}$ is a family of local generators with the strengths $\kappa_i = \|H_i\| + \sum_j \|\Gamma_{ij}\|^2$; with local approximation also with $\kappa_i = \|L_i\|$, and with $\kappa_i = 2\|H_i\| + 2\sum_j \|\Gamma_{ij}\|^2$. Consequently the path-sum bound and its closed forms for $t \ge 0$ (Chapters 2–3) hold for $A(t) = e^{tL} a$: for $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$,*

$$\|[A(t), b]\| \le 2\|a\|\|b\| \Big(\mathbf{1}_{\{X \cap Y \ne \emptyset\}} + \sum_{k \ge 0} \omega_k(t) \Big),$$

where the walk terms $\omega_k(t)$ of (3.6) carry the strengths used: the direct strengths $\|H_i\| + \sum_j \|\Gamma_{ij}\|^2$, or, under local approximation, $\|L_i\|$ or its bound $2\|H_i\| + 2\sum_j \|\Gamma_{ij}\|^2$; the light cone under bounded geometry, and the truncated dynamics of Theorem 5.11.

Proof. G1 is Theorem 5.16, (a) or (b), G2 is Proposition 5.15(c), G3 is the contraction hypothesis; the largest strengths dominate $\|L_i\|$, and G1 is monotone in κ. □

The bounds are for $t \ge 0$. In general a Lindblad family has no contractive negation, and the time-reversed forms of Theorem 3.16 are not available: for the dephasing term of Proposition 5.18 below and an a with $\Gamma a \Gamma = -a$ (such as σ^x for $\Gamma = \sigma^z$), (5.14) at the time $-t < 0$ gives $e^{-t\Delta_\Gamma} a = e^{2t} a$, of norm larger than $\|a\|$. A family without jumps is a Hamiltonian family, whose negation is contractive (Theorem 4.5).

Proposition 5.18 (a dephasing term is contractive). Lean ✓ *For a self-adjoint jump Γ with $\Gamma^2 = 1$ and no Hamiltonian part, $\Delta_\Gamma a = \Gamma a \Gamma - a$, $\Delta_\Gamma^2 = -2\Delta_\Gamma$, and for every real t*

$$e^{t\Delta_\Gamma} a = \tfrac{1+e^{-2t}}{2}\, a + \tfrac{1-e^{-2t}}{2}\, \Gamma a \Gamma; \tag{5.14}$$

for $t \geq 0$ this is a convex combination of a and a unitary conjugate, so $\|e^{t\Delta_\Gamma}\| \leq 1$.

Proof. Such a Γ is a two-valued observable, and the term is pure dephasing in its eigenbasis. With $\Gamma^*\Gamma = 1$ the dissipator is $\Theta - \mathrm{id}$ for $\Theta : a \mapsto \Gamma a \Gamma$, and $\Theta^2 = \mathrm{id}$, so $\Delta_\Gamma^{n+1} = (-2)^n \Delta_\Gamma$. Summing the exponential series, $e^{t\Delta_\Gamma} = \mathrm{id} + \sum_{n\geq 0} \frac{t^{n+1}(-2)^n}{(n+1)!}\Delta_\Gamma = \mathrm{id} + \frac{1-e^{-2t}}{2}\Delta_\Gamma$, which is (5.14). Both id and Θ have norm at most one (Γ is unitary), and the coefficients are nonnegative for $t \geq 0$. □

Example 5.19 (two qubits with dephasing). Lean ✓ Two qubits, $H = \sigma_0^x \sigma_1^x$, one jump $\Gamma = \sigma_0^z$, and the observables $a = \sigma_0^z$, $b = \sigma_1^z$ of Example 4.14. The dissipator vanishes on a, since a commutes with Γ, so $La = \mathrm{i}[H, a]$ and the derivative of $[A(t), b]$ at $t = 0$ is the nonzero double commutator of Chapter 4; hence $[A(t), b] \neq 0$ for some $t > 0$ (the contraction of this term, whose Hamiltonian part does not commute with the jump, is the carried hypothesis, not Proposition 5.18).

For a three-qubit family with random Hamiltonian and jump terms on $\{0, 1\}$ and $\{1, 2\}$ and $a = \sigma_0^x$, $\|[A(t), \sigma_2^x]\|$ is 0 at $t = 0$ and 0.11 at $t = 0.5$; for the same a and $b = \sigma_2^z$ at $t = 0.3$ the commutator norm is 0.026 against the walk bound 4.5 with the strengths $\|L_i\|$ and 272 with the strengths $2\|H_i\| + 2\sum_j \|\Gamma_{ij}\|^2$. Both bounds exceed the trivial bound 2; the first overshoots by a factor of 170, more than the Hamiltonian bounds above, and the larger strengths cost a further factor of 60 (the direct strengths of Theorem 5.16(a) are half of them, and the series they enter is correspondingly smaller).

Notes and references

Local approximation. The review [1], its Proposition 4.1, states for the expectation $\mathbb{E}_S$ onto a region S, the Haar average over the unitaries of the complement of S,

$$\|\mathbb{E}_S A - A\| \leq \sup_B \|[A, B]\| \leq 2\|\mathbb{E}_S A - A\|,$$

the supremum over the unit operators B on the complement. Theorem 5.10 is this statement with $S = X$, $A = M$ and $\mathbb{E}_S = P_X$: the Haar average and the finite twirl over the Weyl unitaries are both the normalized partial trace over the complement, re-embedded (Theorem 5.8), and the supremum is $N_{X^c}(M)$.

Truncated dynamics. The review's Proposition 4.3 bounds the distance from $\tau_t(A)$ to the evolution under the Hamiltonian restricted to a larger region; Theorem 5.11 matches it in shape, an observable of the larger region within the tail of the walk series, with the conditional expectation $P_{X'}A(t)$ as the local approximant and the constants of these notes.

Open systems. The review's Theorem 3.14 is its Lieb-Robinson bound for a local Lindblad generator, for $t \geq 0$, with the norms of the Lindblad terms as strengths. On the qudit net with cyclic alphabets, which has local approximation (Theorem 5.10), Theorem 5.17 with $\kappa_i = \|L_i\|$ and Proposition 3.5 give it in the notation of the review's (3.36) recalled at the end of Chapter 4:

$$\mathcal{C}_{AB}(t) \leq \sum_{u\in A,\, v\in B} \left(e^{2th}\right)_{uv} \quad (t \geq 0), \qquad h_{uv} = \sum_{i:\, u,v\in S_i} \|L_i\|,$$

by the argument given there. The review obtains the contraction of the sub-families from complete positivity; here it is the hypothesis carried in Definition 5.14.

Checks. The twirl against the partial trace (exactly for qubits, numerically for qutrits), the properties of the conditional expectation, the estimate (5.7) with the attained factor 2, the truncation errors on chains of three and five qubits (section 3 of the script; the chapter quotes the five-site numbers), the contractivity, the vanishing off the support, the mechanism $L_S c = L_S(c - P_{S^c} c)$, the strengths and the walk bound for the random Lindblad family (section 4 of the script), and the degenerate cases ($X = \Lambda$, $X = \emptyset$, no jumps, $t = 0$) are computed by `checks/m3_check.py`. The twelve-qubit errors and bounds of Example 5.13 are computed by `checks/long_chain.py`; the fronts, decay rates, ratios and Figure 8 by `checks/long_chain_figure.py`.

6 Fermions

This chapter is to come.

7 Harmonic oscillators

The bounds so far are for bounded local generators: spins and open systems. Harmonic lattice bosons have an unbounded Hamiltonian and an infinite-dimensional algebra, so the generator of their dynamics has no finite strength and the path-sum bound of Chapter 2 does not apply to them. What replaces it is a different mechanism, special to quadratic Hamiltonians: their dynamics moves the Weyl operators along a linear flow of the classical phase space, $\alpha_t(W(f)) = W(T_t f)$, and the Weyl relation turns the commutator of two Weyl operators into a phase, so that the whole question reduces to the entries of the classical propagator $T_t = e^{tK}$.

We bound these entries by the exponential of the entrywise absolute value $|K|$, a nonnegative matrix in the role of the site matrix of Chapter 3: the range form $(vt)^r/r!\, e^{vt}$ and the light cone come over unchanged, with the classical generator in place of the interaction.

7.1 The lattice phase space and the Weyl operators

Throughout, Λ is a finite set of sites. The phase space is $V = \Lambda \to \mathbb{R}^2$, a position coordinate f_x^q and a momentum coordinate f_x^p at every site, equivalently a function on the coordinates $\Lambda' = \Lambda \times \{q, p\}$. For a coordinate $a = (x, q)$ write $\bar{a} = (x, p)$ and conversely, and $\mathrm{site}(a) = x$. The symplectic form is

$$\sigma(f, g) = \sum_{x \in \Lambda} (f_x^q g_x^p - f_x^p g_x^q) = f \cdot \Omega g, \qquad \Omega_{(x,q),(x,p)} = 1,\ \Omega_{(x,p),(x,q)} = -1, \tag{7.1}$$

all other entries of Ω being zero; it is bilinear and alternating, $\sigma(f, f) = 0$. The support of f is the set of sites where a coordinate of f is nonzero, and for $X \subseteq \Lambda$ we write $X' = X \times \{q, p\}$ and $\|f\|_1 = \sum_a |f_a|$.

Definition 7.1 (Weyl system, CCR algebra)**.** A *Weyl system* for σ on a unital C*-algebra $\mathcal{A}$ is a map $W : V \to \mathcal{A}$ with every $W(f)$ unitary, the *Weyl relation*

$$W(f)W(g) = e^{-\mathrm{i}\sigma(f,g)/2}\, W(f + g), \tag{7.2}$$

and the $W(f)$ generating $\mathcal{A}$ as a C*-algebra. The *CCR algebra* of σ is the C*-algebra generated by the concrete Weyl operators on $\ell^2(V)$, V taken with the discrete topology (so $\ell^2(V)$ is not separable); it is the $\mathcal{A}$ of this chapter, with its Weyl system.

From (7.2), $W(0) = 1$ (as $W(0) = W(0)^2$ with $W(0)$ invertible) and $W(f)^* = W(-f)$ (as $W(f)W(-f) = W(0)$). In the oscillator picture $W(f) = \exp(\mathrm{i} \sum_x (f_x^q q_x + f_x^p p_x))$ with $[q_x, p_y] = \mathrm{i}\delta_{xy}$, and the phase in (7.2) is that of the product formula for the exponentials of two operators with a central commutator; the C*-algebraic definition uses the relation and not the picture.

Definition 7.2 (the bosonic net)**.** For $X \subseteq \Lambda$, $\mathcal{A}(X)$ is the C*-subalgebra of $\mathcal{A}$ generated by the $W(f)$ with $\operatorname{supp} f \subseteq X$.

Proposition 7.3 (the form and the net)**.** Lean ✓

(a) If $\operatorname{supp} f \cap \operatorname{supp} g = \emptyset$ *then* $\sigma(f,g) = 0$.

(b) $X \mapsto \mathcal{A}(X)$ *is a net, causal for the lattice index: for* $X \cap Y = \emptyset$*, every element of* $\mathcal{A}(X)$ *commutes with every element of* $\mathcal{A}(Y)$.

Proof. (a) Every summand of (7.1) has a factor from a site outside one of the supports.

(b) Isotony is clear. For $\operatorname{supp} f \subseteq X$, $\operatorname{supp} g \subseteq Y$, Theorem 7.4 below with (a) gives $W(f)W(g) = W(g)W(f)$. The commutant of a set closed under adjoints is a C*-subalgebra, and $\{W(g) : \operatorname{supp} g \subseteq Y\}$ is closed under adjoints ($W(g)^* = W(-g)$); so this commutant contains the generators of $\mathcal{A}(X)$, hence $\mathcal{A}(X)$, and then the commutant of $\mathcal{A}(X)$ contains the generators of $\mathcal{A}(Y)$, hence $\mathcal{A}(Y)$. □

Theorem 7.4 (the Weyl commutator)**.** Lean ✓ *For all* $f, g \in V$,

$$[W(f), W(g)] = \left(e^{-\mathrm{i}\sigma(f,g)} - 1\right) W(g)W(f), \qquad \|[W(f), W(g)]\| = 2\left|\sin\tfrac{\sigma(f,g)}{2}\right| \le |\sigma(f,g)|. \quad (7.3)$$

Proof. By (7.2) twice, $W(f)W(g) = e^{-\mathrm{i}\sigma(f,g)/2}W(f+g)$ and $W(g)W(f) = e^{\mathrm{i}\sigma(f,g)/2}W(f+g)$, so $W(f)W(g) = e^{-\mathrm{i}\sigma(f,g)}W(g)W(f)$, which is the identity. $W(g)W(f)$ is unitary, so the norm is $|e^{-\mathrm{i}s} - 1| = 2|\sin(s/2)|$ with $s = \sigma(f,g)$, and $|\sin(s/2)| \le |s|/2$. □

The commutator of two Weyl operators is thus a number times a unitary, and its size is a function of $\sigma(f,g)$ alone: this is what makes the quadratic case elementary.

7.2 Quadratic generators and the flow

Definition 7.5 (quadratic generator, flow, dynamics)**.** A *quadratic generator* is a real matrix K on Λ' with

$$K^{\mathsf{T}}\Omega + \Omega K = 0, \qquad \text{equivalently} \quad \sigma(Ku, v) + \sigma(u, Kv) = 0. \quad (7.4)$$

Its *flow* is $T_t = e^{tK}$, $t \in \mathbb{R}$. The *dynamics* α_t is the $*$-automorphism of $\mathcal{A}$ induced by the symplectic linear bijection T_t of Proposition 7.6, the one with

$$\alpha_t(W(f)) = W(T_t f). \quad (7.5)$$

The condition (7.4) says that K is a Hamiltonian matrix. For the quadratic Hamiltonian $H = \frac{1}{2}\xi^{\mathsf{T}}M\xi$ in the coordinates $\xi = (q,p)$, M symmetric, Hamilton's equations are $\dot{\xi} = JM\xi$ with $J = \Omega$, and in the Heisenberg picture $\xi(t) = e^{tJM}\xi$. The Weyl operator $W(f) = e^{\mathrm{i}f\cdot\xi}$ therefore evolves into $e^{\mathrm{i}f\cdot\xi(t)} = e^{\mathrm{i}(e^{tK}f)\cdot\xi}$ with $K = (JM)^{\mathsf{T}} = -MJ$: the *label* f moves by the transpose of Hamilton's matrix, which is again Hamiltonian. The generator of this chapter is that transpose, and (7.5) is its defining property; the identification with $e^{\mathrm{i}tH} \cdot e^{-\mathrm{i}tH}$ on Fock space is not part of what is proved.

Proposition 7.6 (the flow)**.** Lean ✓ *For a quadratic generator* K*:* $T_0 = 1$*,* $T_sT_t = T_{s+t}$*,* $T_t^{-1} = T_{-t}$*, and* T_t *is symplectic,* $T_t^{\mathsf{T}}\Omega T_t = \Omega$*, that is* $\sigma(T_tu, T_tv) = \sigma(u,v)$*. Consequently* α_t *exists and is a* $*$*-automorphism with* (7.5)*.*

Proof. The group law is the exponential of commuting matrices. For symplecticity, (7.4) with $\Omega^{-1} = -\Omega$ (from $\Omega^2 = -1$) reads $K^{\mathsf{T}} = \Omega(-K)\Omega^{-1}$, so $(e^{tK})^{\mathsf{T}} = e^{tK^{\mathsf{T}}} = \Omega e^{-tK}\Omega^{-1}$ and $T_t^{\mathsf{T}}\Omega T_t = \Omega e^{-tK}e^{tK} = \Omega$. A symplectic linear bijection T of the phase space induces the automorphism $W(f) \mapsto W(Tf)$ of the CCR algebra: the relabelling $\psi \mapsto \psi \circ T^{-1}$ of $\ell^2(V)$ is unitary, and conjugation by it sends $W(f)$ to $W(Tf)$, hence maps the C*-algebra generated by the Weyl operators onto itself; the automorphisms compose as the maps do. The group $t \mapsto \alpha_t$ is not continuous in norm (Weyl operators with distinct labels are at distance 2); *dynamics* here means the automorphism group. □

One checks that $\alpha_s \circ \alpha_t = \alpha_{s+t}$: both sides are $*$-automorphisms, they agree on the Weyl operators, $W(f) \mapsto W(T_s T_t f) = W(T_{s+t} f)$, and a $*$-automorphism is determined by its values on the Weyl operators, which generate $\mathcal{A}$.

7.3 The entrywise bound

For a real matrix K over any finite index set write $|K|$ for the matrix of absolute values, $|K|_{ba} = |K_{ba}|$.

Lemma 7.7 (powers). Lean ✓ *$|(K^n)_{ba}| \le (|K|^n)_{ba}$ for all n, and $(|K|^n)_{ba} \ge 0$.*

Proof. Induction: $|(K^{n+1})_{ba}| = |\sum_c (K^n)_{bc} K_{ca}| \le \sum_c (|K|^n)_{bc} |K|_{ca}$. □

Theorem 7.8 (entrywise exponential bound). Lean ✓ *For a real matrix K over a finite index set and $t \in \mathbb{R}$,*

$$|(e^{tK})_{ba}| \le \left(e^{|t|\,|K|}\right)_{ba} = \sum_{n \ge 0} \frac{|t|^n}{n!} (|K|^n)_{ba} \ \ge 0. \tag{7.6}$$

Proof. The exponential series converges in norm, hence entrywise, on both sides; termwise $|t^n (K^n)_{ba}/n!| \le |t|^n (|K|^n)_{ba}/n!$ by Lemma 7.7, and the triangle inequality for absolutely convergent series gives the bound. □

Definition 7.9 (range, row sum). Let d be a distance on Λ ($d(x,x) = 0$, the triangle inequality) and site a map from the index set to Λ. K has *range* $R \ge 0$ if $K_{ba} = 0$ whenever $d(\mathrm{site}\, b, \mathrm{site}\, a) > R$, and *row sum* $v \ge 0$ if $\sum_a |K_{ba}| \le v$ for every b.

Lemma 7.10 (range and row sum). Lean ✓ *With range R and row sum v: $(|K|^n)_{ba} = 0$ whenever $nR < d(\mathrm{site}\, b, \mathrm{site}\, a)$, and $\sum_a (|K|^n)_{ba} \le v^n$.*

Proof. Both by induction on n. A nonzero term $|K|_{bc} (|K|^n)_{ca}$ needs $d(\mathrm{site}\, b, \mathrm{site}\, c) \le R$ and $d(\mathrm{site}\, c, \mathrm{site}\, a) \le nR$, hence $d(\mathrm{site}\, b, \mathrm{site}\, a) \le (n+1)R$. Row sums: $\sum_a (|K|^{n+1})_{ba} = \sum_c |K|_{bc} \sum_a (|K|^n)_{ca} \le v^{n+1}$. □

Corollary 7.11 (range form, light cone). Lean ✓ *With range R, row sum v, $t \ge 0$, and an integer $r \ge 0$ with $nR < d(\mathrm{site}\, b, \mathrm{site}\, a)$ for every $n < r$,*

$$\left(e^{t|K|}\right)_{ba} \le \frac{(vt)^r}{r!}\, e^{vt}, \qquad \text{and} \qquad \left(e^{t|K|}\right)_{ba} \le e^{-\mu r} \left(e^{e^{\mu} vt} - 1\right) \quad (\mu \ge 0,\ r \ge 1). \tag{7.7}$$

Proof. Terms with $n < r$ vanish by Lemma 7.10, and the others are at most $t^n v^n / n!$ by the row-sum bound, so $(e^{t|K|})_{ba} \le \sum_{n \ge r} (vt)^n / n!$; this tail is at most $(vt)^r e^{vt} / r!$ by Lemma 3.12. For the cone, $1 \le e^{\mu(n-r)}$ for $n \ge r$ and $\mu \ge 0$, so the tail is at most $e^{-\mu r} \sum_{n \ge 1} (e^{\mu} vt)^n / n!$ when $r \ge 1$. □

Both bounds in (7.7) estimate the tail $\sum_{n \ge r} (vt)^n / n!$, which is itself an upper bound on the entry. For $R > 0$ and $d = d(\mathrm{site}\, b, \mathrm{site}\, a) > 0$ the integer $r = \lceil d/R \rceil$ is admissible; for $R = 0$ and $d > 0$ every r is.

7.4 The harmonic bound

Theorem 7.12 (harmonic Lieb–Robinson bound). Lean ✓ *Let K be a quadratic generator,* $\mathrm{supp}\, f \subseteq X$, $\mathrm{supp}\, g \subseteq Y$ *and $t \in \mathbb{R}$. Then*

$$\|[\alpha_t(W(f)), W(g)]\| \le |\sigma(T_t f, g)| \le \sum_{b \in Y'} \sum_{a \in X'} \left(e^{|t|\,|K|}\right)_{ba} |f_a|\, |g_{\bar{b}}|. \tag{7.8}$$

Proof. By (7.5) and Theorem 7.4, the left side is at most $|\sigma(T_t f, g)|$. By (7.1), $\sigma(u,g) = \sum_b \epsilon_b u_b g_{\bar b}$ with $\epsilon_b = \pm 1$, so $|\sigma(u,g)| \le \sum_{b\in Y'} |u_b||g_{\bar b}|$ when $\operatorname{supp} g \subseteq Y$; and $|(T_t f)_b| \le \sum_{a\in X'} |(e^{tK})_{ba}||f_a|$, bounded by (7.6). □

We state the bound for Weyl operators, as the source does; it extends to finite linear combinations $\sum_i c_i W(f_i)$ by the triangle inequality, with $\sum_i |c_i|$ in place of a norm. A bound for arbitrary elements of $\mathcal{A}(X)$ and $\mathcal{A}(Y)$ in terms of their norms is not claimed.

Corollary 7.13 (closed forms)**.** Lean ✓ *Let K be a quadratic generator of range R and row sum v, with* site(a) *the site of the coordinate a and d a distance on Λ as in Definition 7.9, not assumed symmetric ($d(y,x)$ is read from the site y of the probe); let* $\operatorname{supp} f \subseteq X$, $\operatorname{supp} g \subseteq Y$, $t \ge 0$, *and let $r \ge 0$ be an integer with $nR < d(y,x)$ for all $y \in Y$, $x \in X$, $n < r$. Then*

$$\|[\alpha_t(W(f)), W(g)]\| \le \|f\|_1 \|g\|_1 \frac{(vt)^r}{r!} e^{vt}, \qquad \|[\alpha_t(W(f)), W(g)]\| \le \|f\|_1 \|g\|_1 \, e^{-\mu r} (e^{e^\mu v t} - 1) \tag{7.9}$$

for $\mu \ge 0$, $r \ge 1$.

Proof. Corollary 7.11 in (7.8), and $\sum_{b\in Y'} \sum_{a\in X'} |f_a||g_{\bar b}| \le \|f\|_1 \|g\|_1$. □

For $R > 0$ and $d(Y,X) := \min_{y\in Y, x\in X} d(y,x) > 0$, the integer $r = \lceil d(Y,X)/R \rceil \ge 1$ is admissible in Corollary 7.13, and for $\mu > 0$ the second bound, at most $\|f\|_1 \|g\|_1 e^{-\mu d(Y,X)/R + e^\mu v t}$, is a light cone of velocity $e^\mu v R/\mu$ (at $\mu = 1$: $e\, vR$) and decay rate μ/R in the distance between the supports.

Proposition 7.14 (rescaling)**.** paper *Let K be a quadratic generator, S diagonal with positive entries s_a and $s_{\bar a} = 1/s_a$ (a change of scale $q \to s\,q$, $p \to p/s$ at each site, symplectic), and $K_S = S^{-1} K S$. Then K_S is a quadratic generator with flow $T_t^S = S^{-1} T_t S$; $\sigma(T_t^S S^{-1} f, S^{-1} g) = \sigma(T_t f, g)$, so the commutator of $(K_S, S^{-1} f, S^{-1} g)$ is that of (K, f, g); $|K_S| = S^{-1}|K|S$ and the bound* (7.8) *is unchanged; the row sum of $|K_S|$ is $\max_b s_b^{-1} \sum_a |K_{ba}| s_a$, which depends on S, and the prefactor of* (7.9) *becomes $\|S^{-1} f\|_1 \|S^{-1} g\|_1$.*

Proof. $\sigma(Su, Sv) = \sum_a \epsilon_a s_a u_a s_{\bar a} v_{\bar a} = \sigma(u,v)$ as $s_a s_{\bar a} = 1$, that is $S^{\mathsf T} \Omega S = \Omega$, so S and S^{-1} are symplectic and $\sigma(K_S u, v) = \sigma(KSu, Sv) = -\sigma(Su, KSv) = -\sigma(u, K_S v)$; conjugation commutes with the exponential series and with the entrywise absolute value, since $s_b^{-1} s_a > 0$; in (7.8) the weights $s_b^{-1} s_a s_a^{-1} s_{\bar b}^{-1} = 1$ cancel. □

The entrywise bound is thus invariant under the choice of units at each site, but the range form is not: its velocity is the row sum, which the rescaling can lower.

7.5 The harmonic chain

Definition 7.15 (the harmonic chain)**.** On $\Lambda = \{0, \dots, n-1\}$ with $d(x,y) = |x-y|$, $\omega \ge 0$ and $\lambda \ge 0$,

$$H = \tfrac{1}{2} \sum_x (p_x^2 + \omega^2 q_x^2) + \tfrac{1}{2} \lambda \sum_{x=0}^{n-2} (q_x - q_{x+1})^2 = \tfrac{1}{2} \sum_x p_x^2 + \tfrac{1}{2}\, q^{\mathsf T} A q, \tag{7.10}$$

with $A_{xx} = \omega^2 + \lambda \deg x$ ($\deg x$ the number of neighbours of x), $A_{x,x\pm1} = -\lambda$, and $A_{xy} = 0$ otherwise. Hamilton's equations are $\dot q = p$, $\dot p = -Aq$, and the generator of the labels is their transpose, $(Kf)^q = -A f^p$, $(Kf)^p = f^q$.

Proposition 7.16 (the chain)**.** Lean ✓ *K is a quadratic generator of range* 1 *and, for $\lambda \ge 0$, of row sum $v = \max(1, \omega^2 + 4\lambda)$.*

Proof. $\sigma(Ku, v) = -\sum_{x,y} A_{xy} u_y^p v_x^p - \sum_x u_x^q v_x^q$ is symmetric in (u,v) because A is, so $\sigma(Ku, v) = \sigma(Kv, u) = -\sigma(u, Kv)$. The range and the row sums are read off A, with $\deg x \le 2$. □

One checks from A that the row sums are exactly $\omega^2 + 2\lambda \deg x$ on the position rows and 1 on the momentum rows. The rescaling $s_{(x,q)} = c$, $s_{(x,p)} = 1/c$ of Proposition 7.14 multiplies the position rows by c^{-2} and the momentum rows by c^2, so the row sums become at most $(\omega^2+4\lambda)/c^2$ and c^2, balanced at $c^4 = \omega^2 + 4\lambda$ with $v = (\omega^2 + 4\lambda)^{1/2}$ when $\omega^2 + 4\lambda > 0$ (for $\omega = \lambda = 0$ the position rows vanish, the row sum becomes c^2, and v can be taken as small as wanted).

Example 7.17 (the bound against the exact commutator). script ✓ Twelve sites, $\omega = \lambda = 1$, so $v = 5$; f at the site 0 and g at the site y, both with coordinates of modulus at most 1. Over this class the maximum of $|\sigma(T_t f, g)|$ is attained at sign vectors (a bilinear form on a box), and the bound (7.8) is the sum of the four entries of $e^{t|K|}$ between the two sites. Figure 9 shows both against y. At $t = 0.5$, 1, 2 the exact quantity falls below 0.1 from $y = 2$, 3, 4 on; a least-squares fit of its logarithm over $y = 3, \dots, 10$ gives the decay per site $e^{-6.4y}$, $e^{-5.0y}$, $e^{-3.5y}$, and the same fit of the bound $e^{-6.4y}$, $e^{-5.0y}$, $e^{-3.7y}$ (locally the decay is faster than exponential, as the figure shows).

The bound exceeds the exact value by factors between 1.04 and 1.4 at $t = 0.5$, between 1.1 and 3.4 at $t = 1$, and between 1.7 and 29 at $t = 2$, the largest factors at small y, where the bound also exceeds the trivial bound 2.

The range form (7.9) with $r = y$ and the prefactor $\|f\|_1\|g\|_1 = 4$ is trivial for every y shown at $t = 1$ and $t = 2$, and up to $y = 7$ at $t = 0.5$: at $t = 1$ it reads $3 \cdot 10^3$, $7 \cdot 10^3$, $1 \cdot 10^4$ for $y = 1, 2, 3$ against 1.1, 0.18, 0.0088, because $v = 5$ counts every entry of $|K|$ with its full weight, while the entries of $e^{t|K|}$ between distant sites are small. With the balanced rescaling, $v = \sqrt{5}$, the cone of Corollary 7.13 (the paragraph after it, at $R = \mu = 1$) has velocity $e\,v = 6.1$; the chain's group velocity, the largest slope of its dispersion $\gamma(k) = (\omega^2 + 4\lambda \sin^2(k/2))^{1/2}$, is $(\sqrt{5} - 1)/2 = 0.62$: the cone of the bound is ten times wider than the propagation it bounds.

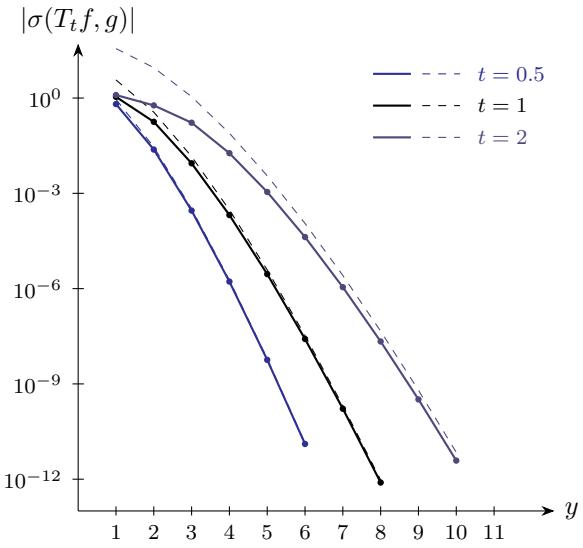


Figure 9: The harmonic chain of twelve sites, $\omega = \lambda = 1$: the exact $|\sigma(T_t f, g)|$ (solid, maximized over f at the site 0 and g at the site y with coordinates of modulus at most 1) and the entrywise bound (7.8) (dashed), against y on a logarithmic scale, for three times. The bound shares the decay of the exact quantity and lies above it by a factor that grows with t.

Proposition 7.18 (witness). Lean ✓ *For $\lambda > 0$ and $n \geq 2$, with $f = \delta_{(0,p)}$ and $g = \delta_{(1,p)}$, there is $t > 0$ with $[\alpha_t(W(f)), W(g)] \neq 0$: the momenta at neighbouring sites do not commute after evolution.*

Proof. $\phi(t) = \sigma(T_t f, g)$ has $\phi(0) = 0$ and $\phi'(0) = \sigma(Kf, g) = (Kf)^q_1 = -A_{10} = \lambda > 0$, so $0 < \phi(t) < 2\pi$ for small $t > 0$, and the norm $2|\sin(\phi(t)/2)|$ of Theorem 7.4 is nonzero. For the

position pair $\delta_{(0,q)}, \delta_{(1,q)}$ the first two orders vanish (Kf and $K^2 f$ pair with g through momentum coordinates only, and $\sigma(Kf, g) = \sigma(K^2 f, g) = 0$) and $\phi(t) = -\lambda t^3/6 + O(t^4)$: the commutator appears at order t^3. □

Notes and references

The source. The harmonic bound is Theorem 3.1 of [7]: for the chain (7.10) on a periodic cube (their Hamiltonian without the factor $\frac{1}{2}$, so their time is half of ours; their chain is periodic where ours has open ends, and in ν dimensions their constant carries $4\sum_j \lambda_j$), with $\operatorname{supp} f \subseteq X$, $\operatorname{supp} g \subseteq Y$ and any $\mu > 0$,

$$\|[\tau_t(W(f)), W(g)]\| \leq C\|f\|_\infty \|g\|_\infty \sum_{x\in X,\, y\in Y} \exp(-\mu\,(d(x,y) - c_{\omega,\lambda} \max(2/\mu, e^{\mu/2+1})|t|)),$$

$$c_{\omega,\lambda} = (\omega^2 + 4\lambda)^{1/2},$$

their eq. (3.10), with $C = 2 + c_{\omega,\lambda} e^{\mu/2} + c_{\omega,\lambda}^{-1}$.

Their proof has the same first two steps as ours, their Lemma 3.4 ($\tau_t(W(f)) = W(f_t)$, obtained there by a normal-mode computation) and the commutator identity (7.3) (their p. 12 and eq. (3.21)); the third step differs: they bound the propagator kernel by a Fourier sum and a power-series tail with an estimate of the factorials (their Lemmas 3.5 and 3.7), while Theorem 7.8 bounds it entrywise by $e^{|t||K|}$ and Chapter 3's tail. Theorem 7.12 with Corollary 7.13 and the paragraph after it gives their statement in shape: exponential decay in the distance with a linear light cone, the constants depending on ω and λ only.

The velocities are not the same number: in their time unit and at the same μ, our cone (the paragraph after Corollary 7.13) moves at $2e^{\mu} c_{\omega,\lambda}/\mu$ after the balanced rescaling, theirs at $c_{\omega,\lambda} \max(2/\mu, e^{\mu/2+1})$; the two agree at $\mu = 2$ and ours is the larger elsewhere, so their cone is the narrower (at the best μ, $3.6\, c_{\omega,\lambda}$ against our $5.4\, c_{\omega,\lambda}$); our prefactor has the simpler form.

The mechanism. That the dynamics of a quadratic Hamiltonian is a symplectic flow on the Weyl labels is classical. The bounded-density bounds for interacting bosons are a different, state-dependent subject and are not treated in these notes.

Checks. The generator condition, symplecticity, the group law, the Weyl commutator norm, the entrywise bound and the power bound, the chain of inequalities (7.8) on a chain of eight sites, the light cone, the row sums before and after the rescaling with the balancing exponent, the momentum-pair witness and the position pair's t^3 onset, the sharper cone tail and the time normalisation against the source are computed by `checks/m5_check.py`; the numbers of Example 7.17 (twelve sites) and Figure 9 by `checks/m5_figure.py`.

8 The thermodynamic limit

This chapter is to come.

Index of terms

Each term is defined where it first appears; the reference is to the definition, or to the statement, equation or section where the term is introduced. Terms marked * are introduced in these notes or used in a narrower sense than usual.

ambient algebra*	§1.1	Lindblad term*	Def. 5.14
approximate cellular automaton*	Cor. 5.12	local algebra	Def. 1.1
boundary ∂Y of a region*	Def. 3.1	local approximation*	Def. 5.1
bounded geometry*	Def. 3.10	local to a region (matrix)*	Def. 4.7
causal net	Def. 1.1	locality in the argument (G1)*	Def. 1.4
CCR algebra	Def. 7.1	net of local algebras	Def. 1.1
clock and shift	Prop. 5.5	path count $P_k(Y, X)$*	Def. 1.7
commutator seminorm*	Def. 1.2	path series*	§2
conditional expectation	Def. 5.3	path-sum bound*	Thm. 2.9
configuration	§4.3	Pauli matrices	Ex. 4.14
contraction (G3)*	Def. 1.4	probe*	§1
contractive negation*	§3.5	probe reset*	§2.3
crossing a region*	Def. 1.4	quadratic generator*	Def. 7.5
decay function*	Def. 3.7	qudit, qudit algebra	§4.3
derivation ad_x	Eq. (4.1)	qudit interaction*	Def. 4.11
dissipator	Def. 5.14	range of a generator*	Def. 7.9
distance*	§3.4	role exchange*	Thm. 4.6
Duhamel's identity	Lem. 2.1	row sum of a generator*	Def. 7.9
dynamics of a family	Def. 1.4	single-site operator	§4.3
embedding $\iota_X m$	Def. 4.7	site matrix*	Def. 3.1
ε-local*	Def. 5.1	strength*	Def. 1.4
F-bound*	Def. 3.7	support closure (G2)*	Def. 1.4
F-norm	Prop. 4.13	total strength*	Def. 1.4
family of local generators*	Def. 1.4	touched sites*	§3.1
first-contact path*	Def. 1.7	twirl	Def. 5.6
flow of a quadratic generator	Def. 7.5	union-chain*	§1
full generator*	Def. 1.4	walk series, walk bound*	Thm. 3.4
Hamiltonian family*	Def. 4.1	walk terms $\omega_k(t)$*	Eq. (3.6)
harmonic chain	Def. 7.15	walk weights*	Def. 3.1
jump operator	Eq. (5.11)	weight of a path*	Def. 1.7
light cone	Thm. 3.14	weighted decay function*	Prop. 4.13
Lindblad family*	Def. 5.14	Weyl operator, Weyl system	Def. 7.1
Lindblad generator	Eq. (5.11)	Weyl unitary	Def. 5.6

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Lean declarations

The linked Lean ✓ tag marks a result whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. Definitions are listed for navigation; only marked statements were reviewed. Files marked (in LinearAQFT), (in LocalAlgebrasLean) or (in BosonicLean) belong to those libraries, on which this one depends, and the declarations listed under them are defined there.

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Chapter 1, Nets, generators and paths	
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	`LocalityLean.LocalGenerators.liebRobinson_abs`
	`LocalityLean.LocalGenerators.liebRobinson_walk_abs`
	`LocalityLean.LocalGenerators.liebRobinson_decay_abs`
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Theorem 4.6	`LocalityLean.Interaction.comm_conj_eq`
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	`LocalityLean.Interaction.liebRobinson_walk_exchange`
	`LocalityLean.Interaction.liebRobinson_walk_min`
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	`LocalAlgebrasLean.Qudit.restrict`
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	`LocalAlgebrasLean.Qudit.IsLocal`
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	`LocalAlgebrasLean.Qudit.embed`
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Lemma 4.8	`LocalAlgebrasLean.Qudit.sum_glue`

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	`LocalAlgebrasLean.Qudit.IsLocal.add`
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	`LocalAlgebrasLean.Qudit.IsLocal.star`
	`LocalAlgebrasLean.Qudit.localAlgebra_isClosed`
	`LocalAlgebrasLean.Qudit.isLocal_embed`
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File `LocalityLean/Qudit/Bounds.lean`	
Definition 4.11	`LocalityLean.Qudit.QuditInteraction`
	`LocalityLean.Qudit.QuditInteraction.term`
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	`LocalityLean.LocalGenerators.fNorm_le_of_isFBound`
	`LocalityLean.DecayKernel.expWeight`
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	`LocalityLean.DecayKernel.expWeight_CF`
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	`LocalityLean.IsApproxLocalWith`
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Definition 5.3	`LocalityLean.Qudit.condExp`
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Proposition 5.4	`LocalityLean.Qudit.isLocal_condExp`
	`LocalityLean.Qudit.condExp_mem`
	`LocalityLean.Qudit.condExp_of_isLocal`
	`LocalityLean.Qudit.condExp_condExp`
	`LocalityLean.Qudit.condExp_add`
	`LocalityLean.Qudit.condExp_smul`
	`LocalityLean.Qudit.condExp_one`
	`LocalityLean.Qudit.star_condExp`
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	`LocalityLean.Qudit.condExp_empty`
Proposition 5.5	`LocalityLean.Qudit.chi`
	`LocalityLean.Qudit.charSum`
	`LocalityLean.Qudit.clock`
	`LocalityLean.Qudit.shift`
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Definition 5.6	`LocalityLean.Qudit.shiftBy`
	`LocalityLean.Qudit.phase`
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	`LocalityLean.Qudit.weyl_apply`
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	`LocalityLean.Qudit.weyl_mem`
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Theorem 5.8	`LocalityLean.Qudit.sum_phase`
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	`LocalityLean.Qudit.norm_condExp_le`
	`LocalityLean.Qudit.condExp_mul_left`
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	`LocalityLean.Qudit.commNorm_compl_le_two_mul_norm_sub_condExp`
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	`LocalityLean.Qudit.QuditInteraction.norm_sub_condExp_evolve_le_cone`
	`LocalityLean.LocalApprox.norm_sub_P_le`
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	LocalityLean.LocalGenerators.evolve_isApproxLocalWith
	LocalityLean.LocalGenerators.neighbourhood
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	LocalityLean.LindbladTerm.gen_apply_mem
	LocalityLean.LindbladTerm.star_gen_apply
Theorem 5.16	LocalityLean.LindbladTerm.jump_apply_eq_comm
	LocalityLean.LindbladTerm.norm_comm_le_norm_mul_commNorm
	LocalityLean.LindbladTerm.norm_jump_apply_le_commNorm
	LocalityLean.LindbladTerm.strengthDirect
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	LocalityLean.LindbladTerm.gen_apply_eq_gen_sub
	LocalityLean.LindbladTerm.norm_gen_apply_le
	LocalityLean.LindbladTerm.strength
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Theorem 5.17	LocalityLean.LindbladFamily.toGeneratorsDirect
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	LocalityLean.LindbladFamily.norm_sub_P_evolve_le
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	LocalityLean.LocalGenerators.liebRobinson_boundedGeometry
	LocalityLean.LocalGenerators.liebRobinson_cone
	LocalityLean.LocalGenerators.norm_sub_P_evolve_le
	LocalityLean.LocalGenerators.commNorm_evolve_le
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Proposition 5.18	LocalityLean.LindbladTerm.jump_eq_of_sq_eq_one
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Files LocalityLean/Approx/Witness.lean, LocalityLean/Qudit/Witness.lean	

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Example 5.19	`LocalityLean.Qudit.TwoQubit.dephasingTerm`
	`LocalityLean.Qudit.TwoQubit.dephasingTerm_gen_obsA`
	`LocalityLean.hasDerivAt_comm_exp`
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Chapter 7, Harmonic oscillators

Files `LocalityLean/Bosonic/Phase.lean`, `BosonicLean/Weyl/System.lean` (in BosonicLean), `BosonicLean/Weyl/Basic.lean` (in BosonicLean), `BosonicLean/Weyl/State.lean` (in BosonicLean)

Definition 7.1	`LocalityLean.Bosonic.Phase`
	`LocalityLean.Bosonic.Omega`
	`LocalityLean.Bosonic.sigma`
	`LocalityLean.Bosonic.sigma_apply`
	`BosonicLean.Weyl.WeylSystem`
	`BosonicLean.Weyl.CCR`
	`BosonicLean.Weyl.wCCR`
	`BosonicLean.Weyl.CCR.weylSystem`

Files `LocalityLean/Bosonic/Phase.lean`, `LinearAQFT/Symplectic/Net.lean` (in LinearAQFT)

Definition 7.2	`LocalityLean.Bosonic.SuppIn`
	`LocalityLean.Bosonic.loc`
	`LocalityLean.Bosonic.phaseSpace`
	`LocalityLean.Bosonic.weylNet`
	`LinearAQFT.Symplectic.ccrNet`
	`LocalityLean.Bosonic.wCCR_mem_weylNet`

File `LocalityLean/Bosonic/Phase.lean`

Proposition 7.3	`LocalityLean.Bosonic.sigma_eq_zero_of_disjoint`
	`LocalityLean.Bosonic.weylNet_isCausal`

File `LocalityLean/Bosonic/Harmonic.lean`

Theorem 7.4	`LocalityLean.Bosonic.comm_wCCR_eq`
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	`LocalityLean.Bosonic.norm_comm_wCCR_eq_sin`
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Definition 7.5	`LocalityLean.Bosonic.IsQuadraticGenerator`
	`LocalityLean.Bosonic.IsQuadraticGenerator.sigma_mulVec_add`
	`LocalityLean.Bosonic.flow`
	`LocalityLean.Bosonic.flowEquiv`
	`LocalityLean.Bosonic.alpha`
	`LocalityLean.Bosonic.alpha_wCCR`
Proposition 7.6	`LocalityLean.Bosonic.flow_zero`
	`LocalityLean.Bosonic.flow_add`
	`LocalityLean.Bosonic.flow_mul_flow_neg`
	`LocalityLean.Bosonic.flow_neg_mul_flow`
	`LocalityLean.Bosonic.IsQuadraticGenerator.flow_symplectic`
	`LocalityLean.Bosonic.IsQuadraticGenerator.sigma_flow`
	`LocalityLean.Bosonic.alpha`
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File `LocalityLean/Bosonic/Banded.lean`

Lemma 7.7	`LocalityLean.Bosonic.abs_pow_apply_le`
	`LocalityLean.Bosonic.pow_absEntry_apply_nonneg`
Theorem 7.8	`LocalityLean.Bosonic.hasSum_exp_apply`
	`LocalityLean.Bosonic.exp_absEntry_apply_nonneg`
	`LocalityLean.Bosonic.abs_exp_apply_le`
Definition 7.9	`LocalityLean.Bosonic.absEntry`
	`LocalityLean.Bosonic.HasRange`
	`LocalityLean.Bosonic.RowSumLE`
Lemma 7.10	`LocalityLean.Bosonic.pow_absEntry_apply_eq_zero_of_lt`
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Corollary 7.11	`LocalityLean.Bosonic.exp_absEntry_apply_le_geom`
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Files `LocalityLean/Bosonic/Phase.lean`, `LocalityLean/Bosonic/Harmonic.lean`	
Theorem 7.12	`LocalityLean.Bosonic.conj`
	`LocalityLean.Bosonic.coords`
	`LocalityLean.Bosonic.abs_sigma_le`
	`LocalityLean.Bosonic.abs_flow_mulVec_apply_le`
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Corollary 7.13	`LocalityLean.Bosonic.liebRobinson_weyl_geom`
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Definition 7.15	`LocalityLean.Bosonic.Adj`
	`LocalityLean.Bosonic.degree`
	`LocalityLean.Bosonic.chainPotential`
	`LocalityLean.Bosonic.chain`
	`LocalityLean.Bosonic.chainDist`
Proposition 7.16	`LocalityLean.Bosonic.chainPotential_isSymm`
	`LocalityLean.Bosonic.chain_isQuadraticGenerator`
	`LocalityLean.Bosonic.chain_hasRange_one`
	`LocalityLean.Bosonic.chain_rowSumLE`
Proposition 7.18	`LocalityLean.Bosonic.exists_comm_alpha_ne_zero`